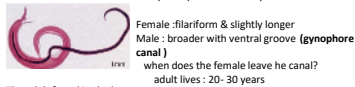
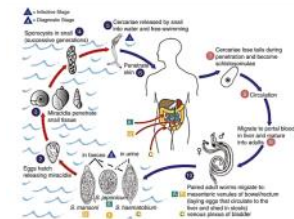


Blood flukes -resides in vessels adult = reside of definitive host here
2 forms intestinal (s.man ,jap.in ,meko) and urinary schistosomiasis (s.hematobium)
Most common in ethiopia : s. mansoni (widely)& s. hematobium What makes schistosome so unique than other trematodes ?bein diecius (separate male & female)
dev. Stage : egg-miracidia-sporozoite - cercaria- adult
Infective for intermediate host (snail) :miracidia -motile (asexually)
Cercaria penetrates the **definitive host** via contaminated water swim till find host if not dies after 3 days
Cercaria shading tail=schistosomula =enter below skin surface = migrates to dermis (10-30 seconds)
Once in circulation --travel to heart/lung/end in veins for liver (grow as adult there)
Adults pair up in these veins, migrate to veins draining guts or bladder, depending on species
s. mansoni: IMV of large intestine
s.hematobium = vesical & pelvic plexus of urinary bladder.



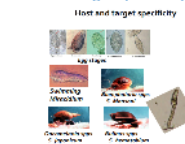
The adult found in the human
1/3 of laid eggs escape body (how ? Penetrating...wall of vein tissues of lumen of...) via passes feces / urine in 50-60 days exposure.....egg hatches[reaching the fresh water - to miracidium (quickly)
miracidium swims till snail host (dies 3- 6 hrs.).... That entering the snail develops to sporocyst(contain rediae)....developes to cercaria[in 4 wks]



Diagnostic stage (pathogenic) :egg (extract from DH)
Infective stage of human: cercaria
Damaging stage :egg stage
- crossing the walls of veins, tissues & mucosa promoting **granuloma formation**
• Eggs deposited in the intestinal wall induce inflammatory reaction

Responsible for continuations of cycle = schistosomula =shaded tail of cercaria

host and target specificity :



pathogenesis & clinical features

Acute (due to cercaria stage) :	migratory phase (4-10 weeks) :heart -->lung -->>	Chronic phase (2 month - yrs) - persons in endemic region got those= ova stage responsible
penetration of skin - swimmers itch ~ 4 days	Toxic / allergic rxn & pulmonary congestion with fever (due to migrated immature stage)	Colicky abdominal pain
abrupt fever (with headache ,shiver, anorexia ,RUQ pain	bronchitis & diarrhea	Dysentery , bloody stool (for yrs)
Less commonly nausea , vomit , diarrhea , cough & bronchospasm		

- clinical complications chronic stage
- Splenomegaly, Ascites (accumulation of fluid in abdominal cavity)
- Hepatomegaly,
- blockage of small vessels
- **Portal vein hypertension syndrome,**
- abdomen distention that looks like a **big drum** (due to enlarged spleen & liver)

laboratory diagnosis

Intestinal schistosomiasis	Urinary schistosomiasis
eggs in feces by direct examination or concentration (formol-ether concentration sedimentation , Kato-Katz technique)	Finding eggs or occasionally the hatched miracidia in urine
Mucus and blood often in feces	Quantify no of eggs/10ml urine by urine filtration technique
Antigen-Antibody detection techniques	Urine contains blood & appears red or red-brown & cloudy.
ELISA, detecting circulating Ag/Ab	Hematuria, Proteinuria, Eosinophilia
 !! S. mansoni: large, triangular lateral spine near the rounded end Morphology of egg - Pale yellow-brown , oval, one well rounded pole & onmore conical pole - Contains fully embryonated miracidium when discharged in feces Specific drug to use : Oxamniquine	 !! S. hematobium: terminal spine at one pole specific drug to use : Metrifonate

Urinary schistosomiasis

- Inflammation of bladder,
 - Hematuria,
 - urinary frequency,
 - pyuria, pyelonephritis,
 - thickening
- ulceration occurs followed by secondary bacterial infection ultimately leading to chronic cystitis
- Obstructive uropathy, cancer of bladder,
- Distant metastases,
 - pulmonary hypertension

Cercarial Dermatitis @ acute stage

- Caused by infectious cercariae of d/t genera
 - Birds: Trichobilharzia, Gigantobilharzia & Bilharziella
 - Mammal: Heterobilharzia genera
- Destroyed by the hosts immune responses,
- Allergic substances released from dead and dying **cercariae** produce a localized inflammatory reaction , Similar but milder symptoms decreasing after a few days. should be differentiated from the swimmers itch.

Prevention, control and treatment

- Traditional or chemical focal molluscicide for snail control
 - Endod (Phytolacca dodecandra) - prof Akiliu Lema
- Proper disposal of urine and feces; Avoid swimming, bathing and washing in snail-infested water
- Provision of safe, adequate water supply, Health information and treatment of infected persons
- Dug : **Praziquantel** effective against all species

- 1. Explain the Schistosoma types with the geographical distributions ?
- 2. What are the acute , intermediate & chronic stages & features related?
- 3. What are the sites for schistosomiasis metastasis?
- 4. More bloody stool and colicky abdomen pain what you suspect?
- 5. What are the damaging factors results granuloma , which phase is more related to this?
- 6. How do we differentiate normal schistosomiasis from cercarial dermatitis?
- 7. Which concentration technique is important for counting?
- 8. Which tests can be used to detect schistosomiasis?
- 9. How do we differentiate the morphology of egg of s.sommani and the s.hematobium ?
- 10. Which drugs are used to treat urinary and the intestinal schistosomiasis ?
- 11. What makes the Schistosoma unique than the other trematodes?
- 12. Explain the cycle of schistosomiasis?
- 13. What is the developmental order for schistosomiasis?
- 14. Which one is infective to snail and human respectively?
- 15. What is the diagnostic agent?
- 16. What is the reason for below you see?



15. which stage characterized by this image?



Human Hydatidosis

Sunday, December 22, 2024 11:25 AM ended @7:45 PM

Definition

zoonotic disease

caused by= Echinococcus granulosus (results ----> intimate contact with dog's feces)

- Two species important :

E. granulosus; causes cystic echinococcosis
(common name: dog tapeworm)

E. multilocularis; causes alveolar echinococcosis
(common name: fox tapeworm)

- highest incidence in sheep raising countries (Australia, Africa, South America & New Zealand)

• Developmental stages: egg, larvae, adult

- Is the **smallest adult stages & the largest larval stage of all tapeworms**

NB: (dead-end host AKA dead-end vector) is an organism that harbors an infectious agent but does not transmit the infection to other susceptible hosts.)

features	Adult tape worm	Larval stage (hydatid cyst)	ova
habitat	lives in the small intestine of definitive hosts (carnivores including dogs, wolves, & coyotes), life span =6-30 months	in various organs of intermediate hosts (Herbivores including sheep, goat and cattle, occasionally infect human that become dead end host - it grows inside IH very slowly	
morphology	3-8mm long • Scolex with four suckers, bears a rostellum with a double row of hooks, neck, 3 segment strobila (immature, mature and gravid) • Segments look like Taenia sp	- The hydatid cyst wall secreted by the embryo consists of 3 layers; pericyst, ectocyst and Endocyst (from its germinal layer small knobs protrude into lumen of cyst , enlarged --> vacuolated --> are filled with fluid :- brood capsules which are initially attached to the germinal layer by stalks, but later are detached and released into the fluid filled cyst cavity. - From inner wall of brood capsules protoscolices develops. Inside mature hydatid cysts, further generation of cyst, daughter cysts and grand-daughter cysts may develop, If a cyst ruptures within a host, each liberated protoscolex can produce a daughter cyst.	round to ova shape --- Embryonated hexacanth embryo inside--- Radially striated egg shell Oncosphere: six hooked embryo inside the egg .

Transmission

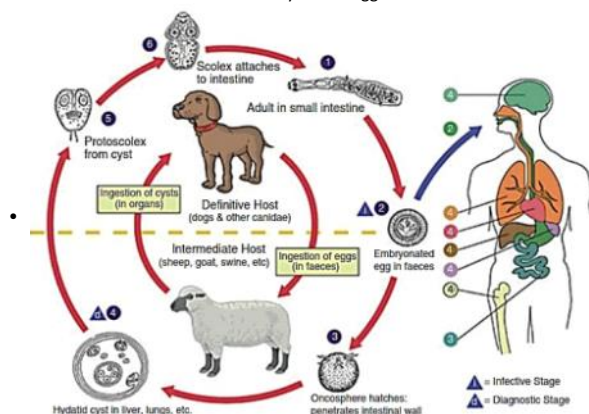
Transmission: ingestion of embryonated eggs

Intermediate hosts (human, sheep, goats, etc.) acquire infection = ingesting embryonated eggs.

- The **egg hatches releasing oncosphere** (penetrates the intestinal wall and **carried by the blood circulation**) Reaches various organs (**liver, lungs, heart, spleen and bones**) where it develops into hydatid cyst.
- Unilocular cyst diameter size = 1-7 cm -----may also be 20 cm /more (human)
- Protoscolices** released from the cyst when **DH ingest hydatid cyst** in organ of **IH** .
- Scolex attachment to intestine led to dev of adult !
- The adult worm of DH sheds embryonated eggs in feces .

Warm up.

- adult is in ? Small intestine
- The infective stage is? Embryonated egg cells .
- Which stage penetrates the intestinal wall and carried out to circulations ? oncospheres
- Where is the hydatid cyst (larval stage) established ? Various organs (liver , lung , heart ,spleen and bones) .
- Which one is the diagnostic stage ? Hydatid cyst
- What follows the cyst? Protoscolex & scolex attaches the intestine
- What is the mechanism of infection of definitive host the IH ? Ingestion of cyst (in organs - DH) but ingestion of the eggs infects of DH is for the IH .



pathogenesis & clinical symptoms

- host tissue rxn **due to** : growing cyst ----leads to-----> fibrous capsule around it!
- obstruction / pressure effect **due to** : grown cyst
- antigenic & its release causing the anaphylaxis is **due to** : cyst filled with fluid
- some got cyst in body 5-20 yrs with /out symptoms /

mechanical blockade & function interference

- Growing hydatid cyst lodges in d/f organs :can you list some ?

- Growing cyst causing obstruction can lead to fatal infection!

Toxic

- Rupture cyst ----cause---->allergy, anaphylaxis & urticaria

Symptoms : dependent on location of cyst .

- liver cyst----> hepatomegaly plus obstructive jaundice : the most common.
- lung cyst----> cough , hemoptysis, chest pain ,pneumothorax & dyspnea
- kidney cyst ---->pain , hematuria
- abd cyst ---> discomfort
- brain cyst ---> ICP high & focal epilepsy

Warm up

1. what happens if cyst ruptures ?
2. where is the position of cyst in case of brain hydatid cyst is occurring ? Above the brain common in child.

Laboratory diagnosis

- examine cystic fluid for (scolice. Hooklets ,brood capsule , protoscolizes)
- X- ray examinations & imaging (usg, CT scan & MRI)
- Serodiagnosis
- **Cason's skin test** (ID screening test)
- Molecular technique (PCR)

Treatment

- Draining cyst (puncture , aspirate , inject, re-aspirate(PAIR) using needle /catheter
- If Cyst accessible surgical treatment
- **Praziquantel** = prior/after OR .
- **Albendazole/mebendazole** = Small / multiple cyst

Prevention & control

- Don't let dog eat carcasses of sheep...
- Deworming dogs periodically
- Personal hygiene
- Dog intimacy area info provisions

actinomycetes

Sunday, December 22, 2024 7:46 PM

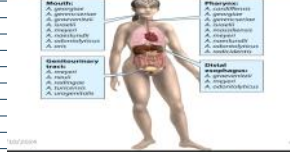
Actinomycetes

- rare - 1/300,000 person cases . M > F = 3:1 ratio (incidence : 20-60 yrs , @peak:40-50 yrs)
 - using IUD (>5 yrs) increases risk for female .
 - increased risk @ rural due to poor hygiene , neglected health status , socioeconomic status
- Chronic
- Slowly progressive
- Granulomatous disease
- Filamentous gram-Positive
- Anaerobic Most are facultative anaerobes except The strictly anaerobic bacteria called *A. meyeri*
- Actinomycetes order
- Actinomycetaceae family
- Genus actinomycetes
- Part of normal flora : oral cavity, GI , female urogenital tract, pharynx , distal esophagus
- Endogenous infection
- access to deep tissue due to trauma/Surgi . procedure/ foreign body disrupting mucosal barrier
- Aggregates of branching , filamentous bacilli inside tissue



electron micrograph of *Actinomyces israelii*, the leading cause of actinomycosis causes abscesses on the face and neck.

Natural habitats of human Actinomycetes species

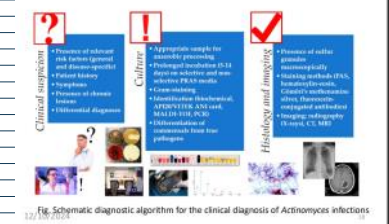


Diagnosis

- **Confirming diagnosis : sterile site identification**
- Isolation & identification in minority cases factors can be: **antibiotic therapy previously** , inhibited growth due to contaminant microbes , inadequate culture , inadequate short incubations (the bacteria needs 10 days):
- nb : need strict anerobic processing & growth conditions .culture needs incubated @ anaerobic atmosphere .
- most appropriate specimen is : tissue biopsy /pus - & gram stain is more sensitive .

do not use swab!!

- prolonged culture (slow growth @least 5 days , takes 15-20 days with IP of 10 days) - in appropriate media (chocolate bold agar media , brain heart infusion broth & brucella blood agar with hemin and vit K) & app atm !
- most actinomycetes species are facultative anaerobes .



pathophysiology

- 1. **low grade pathogen** : not possess classical virulence factors such as exotoxins .
 - causes disease (if mucosal barrier disrupted by exogenous /iatrogenic damage. Can you list some please?
- 2. **chronic lesion** : evades elimination by the immune cells .
 - : dense , interlinked chain of branching bacteria in **Sulphur granules** , macroscopically - acts as inhibitor of phagocytic clearance .
 - start @ 5:23 am
 - infection is poly microbial (involves simultaneously 5-10 other bacteria) - aerobic /facultative enhance survival of anaerobes via
 - reducing o2 tension in tissue ---->destruction Tissue provide nutrient & fe+2 ions----> presence of foreign body increases invasiveness..
 - Producing capsule /toxins
 - Damage mucosal barrier
 - Enhancing invasiveness of actinomycetes species

- are isolated with other normal commensals depending on site of infection .can you list some?

histopathology: Sulphur granules

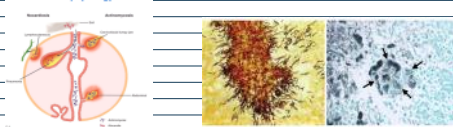


Fig. Sulfur granule formed for its gross resemblance to a grain of sulfur is a microcolony of combined branching Actinomyces filaments embedded with elements of tissue reaction.

Major clinical manifestations

clinical manifestation	Prevalence	clinical cxs
Cervicofacial (including CNS) highest in lower & upper mandible (50%) lowest in tongue/sinus/middle ear./larynx , thyroid gland , lacrimal pathways (0-5%) cheeks and chin same %= (10-15%) CNS & submaxillar ramus ,angle ,mandibular joints got same % = (5-10%)	40 -60 % (1st comm on site)	> The Most common site . > Risk factors = Poor hygiene ,medical interventions , trauma to mouth /jaw . > Swollen indurated appearance when lesion is on : submandibular region & angle of jaw
Abdominal	20-30 % (2nd site)	● as consequence of invasive procedures /Abd infection (e.g. Appendicitis) ● <i>A. israelii</i> ----> common infectious species ● Appendix ,cecum ,colon ----> common sites !
thoracic (inc. pulmonary)	20 -25 % (3rd)	→ Includes pulm ,trach ,laryngeal actinomycetes → occurs in lung & airways → 3rd most common site → Aspiration of oropharyngeal/gastrointestinal secretions → Risk factors = poor oral hygiene , preexisting dental disease , alcoholism , chronic lung death.
Pelvic	3-5 %	⊗ genitourinary - mostly in women - actinomycetes spread from genital to pelvis . ⊗ Can be due to long term use of IUDs .
Cutaneous	3-5%	

Treatment

- Antibiotics
 - Penicillin G (if allergic clindamycin & macrolides(such as erythromycin ,clarithromycin /azithromycin) can be used) , doxycycline
 - Beta lactams (long course oral amoxa)
 - Cephalosporin
- ,surgery(adjunctive)
- Polymicrobial infection : use metronidazole + beta lactamase inhibitor , duration is 6-12 month ,if surgery carried out the duration is decreased.

Nocardiosis

Monday, December 23, 2024 1:17 PM

About

- Many strains => poor for gram stain
- Appears "beaded"- filamentous
- Alternating gram positive & negative of same filament
- Nocardia species ubiquitous in env't (soil)
- Organisms isolated in small number from respiratory tract of healthy (but not considered as member of microbiota)
- Fully developed colonies of species -----gives aroma ----> wet dirt
- Species most common to human : N. Asteroids & N. Brasiliense's = weakly acid fast
- Global incidence not well known ...50-1000 cases /year(USA)
- Affects all ages & races (more in males = 3:1 ratio)
- 60% of pts seen were with immunocompromised status too.
- 33% pts are immunocompetent (cutaneous infection)

Summary

Videos	sketchy	physio	Pixorase
			
			Nocardia-@USMLE...

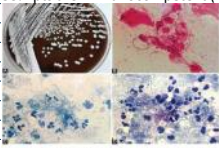
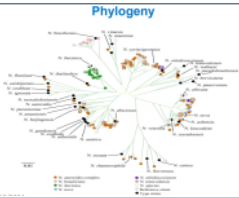


Fig. (A) Colonies of Nocardia on blood agar plates. (B) Clinical isolates with branching Gram-positive bacilli. (C) Clinical isolates with negative acid-fast staining. (D) Clinical isolates with positive partial modified acid-fast staining.

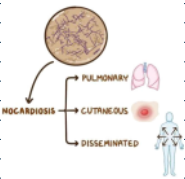


Risk factors

- Immunocompromised pts
- HIV (most common)
- Cancer
- Chemotherapy
- Pts on Corticosteroids
- Lung transplant recipients
- DM
- Autoimmune disease
- Chronic lung disease (COPD)
- **immunity** (dominant host defense = effective T-cell mediated immunity So pts with impaired immunity greatest risk) **NB**: little evidence of humeral immune response.

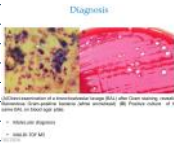
Pathophysiology

1. cutaneous disease ((localized infection with abscess and ulcer) :
 - acquired via local injury to skin (e.g. injection by thorn pricks.)
 - not serious on immunocompetent
 - spreads to other organs in the compromised one
 2. pleuropulmonary nocardiosis
 - inhalation of aerosolized droplets of bacteria.
 - mainly in compromised pts due to immunosuppressive therapy.
 - transplant pts are prominent representative
- NB** no case-case transmission



Diagnosis

- Direct examination of a bronchoalveolar lavage (BAL)
- Molecular diagnosis
- MALDI-TOF MS



Treatment

- Sulfonamides (since 1940s): Trimethoprim-sulfamethoxazole (TMP-SMX) => most common

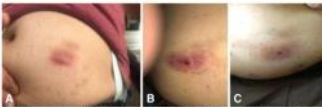
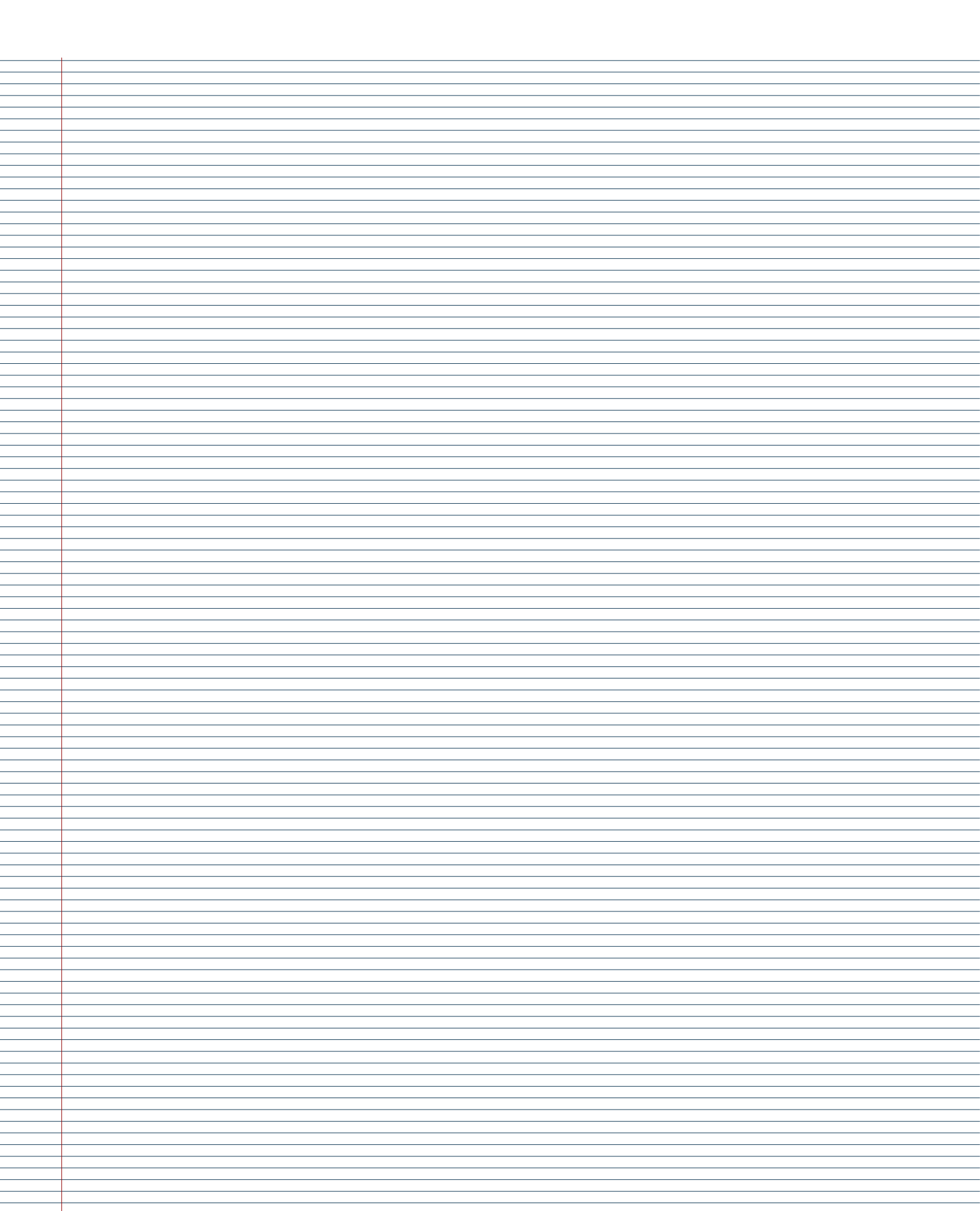


Fig. A Red erythematous papules (4*3 cm) of cutaneous nocardiosis. B. Erythematous papules forming the plaque with erosions, pus discharge, and crusting. C. The erythematous plaque reduced and liquified after successful treatment with TMP-SMX.

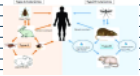
- Other agents
 - Carbapenems : imipenem /meropenem /ertapenem
 - Quinolones : moxifloxacin
 - Tetracycline : minocycline
- Duration of treatment depends on => location & immune status of pts.
 - Pulmonary/multifocal disease without CNS = 6-12 month (~1 year)
 - Immunosuppressed / CNS involved = at least 1 year.



- Zoonotic infection
- Attacks the lymph nodes , skin ,lungs , & eyes
- Also rabbit fever / deer fly fever
 - Most in USA , rare (Africa)
 - Also in Europe(turkey) , Asia (mississippi river & pacific northwest)

- **Francisella tularensis**
 - (i) Gram negative
 - (i) Very small
 - (i) Cocci/bacilli
 - (i) Thin lipid capsule
 - (i) Non-motile
 - (i) Aerobic; facultative anaerobe, intracellular
 - (i) Cysteine for growth
 - (i) Isolated from McCoy & chajin -Tulare country(California -1911)
 - (i) Characteristic & definition by Edward Francis

1. Arthropod vectors bite - ticks , deer fly , mosquitoes
2. Direct contact - infected /dead animals
3. Via aerosols ; bioterrorism
4. Ingestion of contaminated h2o , food / soil

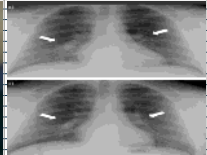


- Risk factors**
- Occupational - veterinarians - LAB professionals - farmer , hunters , butchers ; hikers

- **Intracellular**
 - highly virulent (only 10-50 organism → disease
 - localize-intracellular (inside macrophages & neutrophils); it reproduce within them
 - inhibits **phago-lysosome** functions
 - bacterial release via cell lysis
 - pill , capsule , & uPS too are virulence factors .



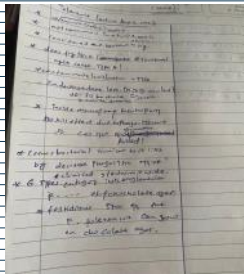
- **Clinical disease**
 - wide spectrum sign & signs
 - mild to life threatening illness
 - bacterial species & portal of entry matters : **tularemia** is sever
 - abrupt symptoms occur
 - IP -3-5 days
 - non-specific
 - May occurs systemic manifestations

Forms of tularemia					
Ulcer- glandular	glandular	Oculoglandular	Oropharyngeal	pneumonic	Typhoidal
• Most common	• Second most common	• Conjunctival infection	• Direct ingestion of bacterial contaminated food /water	• Most serious form.	• Systemic-febrile illness (blood stream & organs)
• Skin lesion + lymphadenopathy	• Tender regional lymphadenopathy	→ Unilateral eye symptoms	→ Mouth & throat	• Primary & secondary	• Symptoms
• Cause :inoculation of skin (tick bite)	• No skin ulcer	→ Pain	→ Fever	1. Primary	→ Fever
		→ Photophobia	→ Sore throat	→ Direct inhalation of bacterium(farmers , lab workers)	→ Chills
		→ Increased tearing	→ Swollen cervical lymph nodes .	2. Secondary	→ Myalgias
		→ Tender regional lymph adenopathy	→ exudative pharyngitis	→ Hematogenous spread to lung.	→ Malaise
				• Symptoms	→ Weight loss
				→ cough	→ Diarrhea
				→ Fever	• Severe
				→ Chest pain	• Absence of lymphadenopathy.
				→ Bilir -adenopathy	
					
					

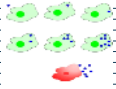
- **Diagnosis**
 - Suspected- based on symptomatology -sign ,symptom & history
 - Laboratory : microscopy sensitive
 - Culture : sensitive + specific ; **cysteine supplemented media** → chocolate agar ; slow growth ; biochemical test
 - Serology :4X increase in titer/ single titer ≥1:160
 - False positive (cross reactive with brucella)

- **Treatment**
 - Aminoglycosides (strep and gentamycin)
 - Fluoroquinolones (cipro & doxycycline)

- **Prevention & Control**
 - avoid reservoir / vectors of infection
 - live attenuated vaccine available (rarely used for human disease)



The class audio



Facts	Embryology	Facts	Agent	Virulence	Source of infection	Transmission	Risk groups	Clinical disease
• Opportunistic infection in human • Cause: <i>Burkholderia pseudomallei</i> - named after the microbiologist • Clinical & pathological similar to glanders disease - different by ecology & epidemiology • Endemic - tropical & subtropical region of world - discovered in Australia • Epidemiologic infection (sheep, horse, swine) • It carries the high mortality • As biological weapon => its strains found in the environment • It is easily neglected • Multiple antibiotics resistance • No vaccine Notes  → Small → Motile → Aerobic, gram negative → Non fermentative → Facultative intracellular → Strains are divided into 14 → Related to <i>Burkholderia mallei</i> Example  • Environmental bacteria • Soil & muddy water • Polluted (contaminated) rain spread • animal doesn't fall in interval but infected • human-human transmission possible (contact) Transmission → Contamination of skin abrasion → Ingest contaminated H2O → Inhalation of dust (for born transmission) → Zoonotic potential (few case) Risk group • Immunocompromised - HIV, DM, chronic renal /lung disease • Chronic alcoholics Virulence & pathogenesis → Facultative intracellular - invade tissues, resist factors of serum, survive intracellular Clinical disease	Pasture	disorder glanders as a Biological Weapon → History • World War I - infected Russian horses and mules → World war II - Japanese horses → World war II - Japanese horses → Prefer media with glyceral → Medical use as vaccine → Very few organism is required → In case of death → Easily produced → Airborne transmission → Limited experiments with disease • can slow diagnosis and treatment → Lack of an effective vaccine	• Small • Motile • Gram-negative • Non-fermenting • Facultative intracellular • Grow on standard medium • Related to Burkholderia mallei • Epidemic - regional & subregional region of world - discovered in Australia	• Facultative intracellular pathogen • Ability to invade tissues, resist factors in serum, & survive intracellularly	• Environmental bacteria • Soil & muddy water • Polluted /contaminated atm spread • animal doesn't act as reservoir • Not infected • human-human transmission possible (contact)	→ Contamination of skin abrasion → Ingest contaminated H2O → Inhalation of dust for born disease → Chronic atelectasis	• Immunosuppressed - HIV, DM, chronic renal /lung disease • Chronic alcoholics	Cutaneous infection Forms systemic infection Can spread to lungs, myocardium, liver & bone Pulmonary disease Most dangerous infection form Upper lobe of lung Progressive illness cause cavity mimics

	
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Diagnosis

→ Suspected based on epidemiology, sign, symptom, history

Lab diagnosis

→ Microscopy when grow faster - near gram negative, specific bipolar staining

→ Culture on **Alabaster's medium** (selective medium)

→ Sensitivity to agglutination test (striped disk or tube agglutination), ELISA



Treatment

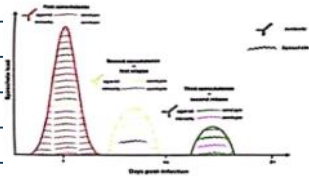
- Surgical drainage
- Drugs
 - Ceftriaxone (highly effective)
 - Tetracycline, aztreonam, sulfamethoxazole
- **Prolonged treatment (1-6 weeks & 2 wks)**

	Diagnosis & Lab	Treatment	Images
	• Suspected based on symptomatology • Look for sign and symptoms, and history • Direct microscopy: gram staining • Grow out on blood agar • Microscopic sign: non-reactive • Large bacillary colonies with fleecy odor • Biochemical tests: rapid • MIC: catalase: weak catalase neg. • Oxidative: biochemical: weak catalase positive	• Susceptible to a variety of antibiotics • Penicillin = 1st choice • Alternatives • expanded spectrum • cephalosporins, macrolides, tetracyclines, or fluoroquinolones	
	• Suspected based on symptomatology • Look for sign and symptoms, and history • Chest radiograph for the pulmonary form of disease • Culture and Gram stain • Sputum, urine, sput. lesions, blood • Gram negative bacilli • Safety pin appearance • Serology • Agglutination tests • May be positive after 7 to 10 days • Complement fixation • More specific - fraction of test is equal four greater than 1:32 • PCR	• Limited information on treatment • Long term antibiotic treatment necessary (1 to 12 months) • Multiple drug therapy • Drain abscesses	
MM	• Suspected based on symptomatology • symptomatology sign • symptomatology history		
why rhinos	• Lab diagnosis • Microscopy after gram stain: weak gram-negative, specific • Agglutination • Culture → Ashdown's medium (selective medium) • Serology → agglutination • Indirect slide or tube agglutination (1:16)		

Relapsing fever

Tuesday, December 24, 2024 4:11 AM

- Recurring fever disease
- Transmit by ticks (endemic one) : soft ticks genus ornithodorus
- Cause = several different types of Borella species @ least 15 species
- **Endemic** relapsing fever causing species named acc to species of tick --> transmitting bacterium to human
e.g. B.parkeri transmitted by o.parkeri , B.HERMSIL (by o.hermsi)



- IP of TBRF = 4-18 days ---then---> 12 recurring febrile episodes (2-7 days; fever & 10 days ;afebrile)
- Primary febrile episode----culminates in ----fatal shock.
- Systemic complications
 - Nausea & vomiting , Upper Abd pain (due to liver and spleen involved) , dry cough
- other complications
 - Splenomegaly , hepatomegaly , jaundice , rash , respiratory symptoms , CNS involved

Pathogenesis

- ♦ Endemic RF : soft ticks feed for short periods & fall off host ;during night
- ♦ Pt infected with borrelia within minutes after tick attaches to host VS ticks for Lyme disease : feed for longer periods (several hrs. for borrelia burgdorferi)
- ♦ Spirochetemia (large no in blood stream) can (highly variable) 1:104 generation the Borella invades epithelium .
- ♦ Multiply in epithelium Cells => bacteria released to stream => bacteria develop from host => symptom resolved when Abs secreted to proteins of outer surface of bacteria from Stream.
- ♦ But Borrelia in endothelial cell (not killed , changes the protein on its outer surface)

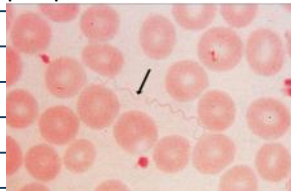
Epidemiology of Borrelia Infections

	Infection	Reservoir	Vector
Borrelia recurrentis	Relapsing fever Epidemic (louse-borne)	Humans	Body louse <i>Pediculus humanus</i>
Borrelia spp.	Relapsing fever Endemic (tick-borne)	Rodents, soft-shelled ticks	Soft-shelled tick <i>Ornithodoros</i> spp.
Borrelia burgdorferi	Lyme disease	Rodents, deer, domestic pets, hard-shelled ticks	Hard-shelled tick <i>Ixodes</i> spp.

Diagnosis

Blood smear from febrile patient

- Wright/Giemsa/Leishman stain
- Microscopic examination: Observe spirochetes in the peripheral blood smear.



Giemsa stain: Tightly coiled helical spirochetes of *Borrelia recurrentis* in the blood of a patient with relapsing fever.

Treatment : Relapsing fever

- o Drugs of choice: Tetracycline/Penicillin/Erythromycin/CAF
- o JHR (Jarisch-Herxheimer Reaction): 1-2 hrs after Rx
- o MR: 1-5%
- o Endotoxin released from dead spirochetes stimulates the release of cytokines (IL-6, TNF alpha, IL-8)
- o Hypotension, shock, death

Fever symptoms



Prevention

- o Personal hygiene: Louse borne RF (more sever CNS involvement)
- o Ticks/rodents control: Tick borne RF (less severe)
- o No vaccines have been developed for RF

Epidemic typhus

1. endemic typhus

- Cause = **rickettsia prowazekii**
 - By **body louse** = pediculus humanus corpus
 - flying squirrels and ticks, which can serve as reservoirs and sources of infection.
 - Oldest pestilential disease
 - disastrous = outcome of wars ; identified in Paris (1083)....16th century (in Italy)
 - Charles Nicolle
 - AKA
 - epidemic typhus decreased after WW II)
 - louse born typhus ,
 - exanthematic typhus
 - historical typhus ,
 - sylvatic typhus
 - calssic typhus (reemerged @ WW I)
 - red louse disease
 - jail fever : it is endemic in Ethiopia since 1866 - prisons & army campus)
- Since 1981 it emerged on highlands of Africa , central & south America

Clinical manifestations

- Rash (rare in Africa ;20 -40 %)
- Nausea & vomiting (42% - 56%)
- Coughing & pneumonia (38% -70%)
- rads of symptoms include fever, headache and rash
- CNS function abnormalities
- Myocarditis
- Pulmonary involvement
- Thrombocytopenia (mild)
- Jaundice In severe cases
 - uncomplicated typhus , fever lasts 2 weeks (untreated) -----recovery (2-3 month)
 - if no treatment = fatal (13-30%)



2.murine typhus (endemic typhus)

- Cause = **rickettsia typhi**
- By **rat flea** =Xeno psylla cheopis , also the c. Felis (check it out)
- **Urban rats** is reservoir
- Human illness is **incidental**
- similar Pathogenesis to epidemic typhus
- Clinical manifestations**
 - Flea defecates when take blood meal (feces get in via bite wound)
 - IP = 1-2 weeks .
 - illness begins with
 - headache
 - myalgia
 - fever
 - Ag shared by the two disease = similar
 - Untreated patient fever =12-14 days (2 weeks)
 - If No treatment = rare mortality & complications

3.scrub typhus

- Causative agent = **orientia tsutsugamushi** , rickettsial organism
- Mites are vectors & reservoirs (those infecting rodents)
- Transmits rickettsiae to their own progeny via **infected ova**
- Human picks mites as passing by low trees/brush
- Chiggers (mite larvae)=>deposit rickettsiae as they feed .
- **Scrub typhus triangle** (China ,Indonesia , south Asia)
- Geographic range is expanding (India ,Micronesia , Maldives ,Russia , Japan)
- Clinical manifestation**
 - ❖ Typical initial lesion = **necrotic eschar (50 % -80 % of cases)**
 - ❖ Slow increasing **fever** over 1st week = 40.5°c
 - ❖ Symptoms after
 - Headache
 - Rash
 - Generalized Lymphadenopathy
 - Hepato-splenomegaly & conjunctivitis (may occur)
- ❖ Untreated pts =mortality (30%)

Diagnosis

- ❖ Microbiological DX = mainly => serology
- ❖ Additional tests = plate microagglutination + indirect IFT .
- ❖ Ag similar to the formers
- ❖ Molecular technique = PCR => detects rickettsia from blood & lice.
- ❖ Differential diagnosis =based on the increasing titer
 - acute or convalescent serum samples (epidemic typhus) ; >4X

Interpretation of Weil-Felix test

	Ab-OX19	Ab-OX2	Ab-OXK
Epidemic typhus	++	±	-
Endemic typhus	++	-	-
RMSF	+	+	-
Scrub typhus	-	-	+
Q fever	-	-	-

Treatment

- ☎ Tetracycline (doxycycline)=> effective , daily orally & 3-4 days deference in epidemic typhus .
- ☎ The course of fever reduced to 2-3 days
- ☎ Chloramphenicol (can be effective)
- ☎ DON'T give sulfonamide = enhanced the disease
- ☎ Fluroquinolones (in vitro activity) but limited clinical experience

Prevention

- Prevention of louse infestations (for epidemic typhus)
- Delousing methods ; what can be done during the process?
- Insecticides (10% DDT , 1%malathion / 1%permethrin .) = louse eradication but eradication of the disease is impossible (retains the R.prowazeki for life time)

Plague

Tuesday, December 24, 2024 1:09 PM

- Term = used generically describing explosive pandemic disease (with high mortality)
- Medically refers infection
- Caused by : **Yersinia pestis** = cause of most virulent epidemic plague ever recorded (**black death**)
- 14 th C _25 M death (1/3rd) _Europe
- Characterized by **swollen lymph nodes (buboes)**=deadly epidemics
- Humans are **incidental** , "DEAD-END " host

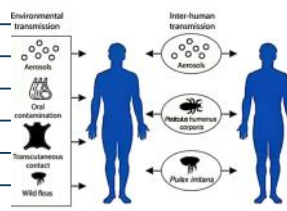
About :Yersinia pestis

- Non motile
- Gram negative bacillus/coccobacillus
- **Hairpin appearance** after gram staining
- **Striking bipolar staining** in special stains (wright ,Giemsa ,Wayson /methylene blue)
- Is pathogenic Yersinia species (like Y. Enterocolitica & Y. pseudoTB)
- Easily isolated when(**blood and lymph aspirates** plated on sheep blood agar)



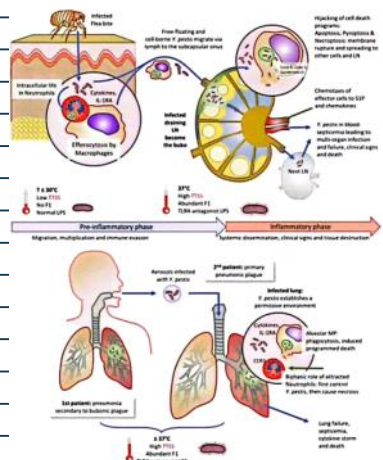
Transmission

- ◇ by plague bacillus transmitted via : flea bite , exposure (to fluids/tissues of infected animal)
- ◇ High mortality (40% -100%)
- ◇ Potent **biowarfare**



Yersinia pestis pathogenesis

- ▶ All possess LPS (endotoxic activity)
- ▶ Y. pestis & Y. enterocoli produce antigens & toxins = virulence factors
- ▶ Type III secretion systems (allows it to inject proteins into cytoplasm)
- ▶ Virulent Yersinia got V & W antigens (plasmid);essential for virulence => yields the requirement for **Ca+2 for growth @37 *c**
- ▶ Additional plasmid (PPCP1) =>contains genes of plasminogen activating proteases .



[illegible]

- ◇ Depends on route of exposure
- ◇ 3 forms : bubonic (most common presentation) , pneumonic plague , septicemic plague

If untreated = 50%(mortality)
Pneumonic plague =100%(mortality)

- streptomycin (nephrotoxic & ototoxic **caution use** (pregnant , elders ,children)
- aminoglycoside= gentamycin is more effective
- Combination of the two .

fascioliasis

Tuesday, December 24, 2024 6:59 PM

- Water born/food born zoonotic disease
- Cause : class Trematoda , genus fasciola
- 1. F. Hepatica (AKA liver fluke/sheep liver fluke)
- 2. F. gigantica (AKA large liver fluke)
- What other liver flukes you know ? C. sinensis & O. viverrini ; donot cause fascioliasis
- Cosmopolitan distribution (prevalent in sheep & cattle raising countries)
- Take breath , it is NOT in Ethiopia
- Human not primary host ! (but got economic crisis)
- Few cases (<2%)=> eggs in stools who consumed infected liver.
- Development stage : **egg (in feces) => miracidia (in fresh water snail) => sporocyst (in snail) => rediae (in snail) => cercaria => metacercaria (infective stage it's on water vegetations) => adult (liver/ biliary duct)**

Fasciolopsis buski, also known as the giant intestinal fluke, is the largest intestinal fluke that infects humans. Humans get infected by ingesting metacercariae (encysted larval stage) on aquatic plants like water chestnut and water caltrop.

Paragonimus westermani, commonly known as the oriental lung fluke, infects humans by eating inadequately cooked or pickled crustaceans (like crabs or crayfish) that harbor metacercariae

Adult	Eggs
Fleshy ,flat ,leaf-like ,gray-brown ,large hermaphrodite 30x12mm	Large Underdeveloped (immature)ovum (are excreted)/ yellow - dark brown
Cone shaped prominent shoulder	Smooth shell(with double layer)
● In liver/biliary duct of sheep/cattle /goat /man	Oval with rounded poles
	Marked operculum @one pole
	130-150 um by 60 -90 um
	Bile stained
	eggs embryonated in H2O (in 4-15 days)=>releasing miracidia .
	● In feces

he condition not due to microfilaria is onchocercoma.

Onchocercoma is a nodule formed under the skin due to the adult worm of Onchocerca volvulus

Transmission

- Mammals = eating vegetation containing **metacercaria**) => it **excyst in Duodenum** & migrates to peritoneal cavity via intestinal wall .then penetrates liver capsule & reaches biliary ducts (develop as adult here)
- metacercaria to adult (maturation takes 3-4 months)
- **Miracidia invades snail intermediate** (in 8 hrs.)
- Cercariae released from snail => encyst as metacercaria on aquatic vegetations .

exte

Pathogenesis & clinical features

- ❖ Inflammatory rxn evoked by : worms and their metabolic products
- ❖ Larval migration through liver got effects
 - Parenchymal injury
 - Fever
 - Headache
 - Chills
 - RUQ pain
 - Eosinophilia & hepatomegaly
- ❖ Larvae encyst & calcified (while undergoing ectopic migration to reach lung, sub .cu tissue , genitourinary tract & brain)
- ❖ Chronic infection
 - mechanical destruction of liver tissue & bile duct & obstruction , obstructive jaundice
 - biliary cirrhosis
 - diarrhea
 - hemorrhage & anemia
 - portal vessel atrophy

laboratory diagnosis

Serology (abs from serum)	Concentration techniques(few eggs found in aspirate & bile)
Eggs in feces (chronic infection) , its needs confirmation of true infection e.g. (ingesting animal liver with egg can false positive , keep him free of diet then after 3 days checked up)	Imaging CT/ us
embryonic egg = diagnostic stage.	

treatment

- triclabendazole or

Prevention

- Prevent pollution of (water)
- sanitation & disposal of human feces
- Wash water plants or cook well
- Identifying & destroying snail hosts & snail habitat
- Fencing land with metacercaria

Trichinosis

Tuesday, December 24, 2024 9:56 PM

- AKA trichinellosis ,trichiniasis , trichinelliasis
- Zoonotic disease
- Cause= T. Spiralis
- In pork eating countries (Europe , America , Africa)
- DH=IH (same host act as both)
- Human =incidental host , pigs & rats = reservoir host (infected thru eating dead/live flesh of others -cannibalism)
- One of smallest nematodes inf human => with very short life span
- Male worm (dies after fertilizing female) , female (viviparous - deposits larvae @ intestinal mucosa=> larva enter blood => to go to striate muscle and encysted) dies (after discharging larvae=2 month

Morphology

Male	Female	Encysted larva
1.5 x0.04 mm(L)	3.5 x0.06 mm (L)	cystic wall(by calcified tissue reaction)
Post. End curved ventral	Post .end bluntly round, vulva opens @anterior 5th	becomes infective in 2 wks. Encystation is carried out @striated muscles) ; it coils & encyst in long axis of muscles .

- Adults =in small intestine of man/pig/rats => the larvae released to intestine => penetrates mucosa (develops as adult)
- Larvae =encyst in straited muscle =coiled 2 & half x

Transmission

- Pork flesh with encysted larvae
- Distributed all over body(tongue , deltoid ,pectorals ,diaphragm....etc.)

Pathogenesis

Intestinal invasion	Migration of larvae	Encystment of larva
• By liberated larva /adult	◇ Fever	◇ Depends on the organ
• Abd pain	◇ Facia oedema	◇ Myocarditis +encephalitis (severe ;can be fatal)
• Vomit	◇ Periorbital swelling	
• Diarrhea	◇ Myalgia	
• colic	◇ Weakness of affected site	

Diagnosis

Immunodiagnosics (intradermal test = batch man test)

Xenodiagnoses

Muscle atrophy (direct exam on larvae , after digestion in pepsin HCL media)

Xray (calcified cyst &PCR)

Treatment

Supportive TX with rest, fluids, smooth diet and vitamins

- Analgesics, antipyretics, sedatives, Cortisone, steroids
- Albendazole, Mebendazole

Prevention and control

- Inspection of pork in slaughter houses and proper cooking

- Do not feed pigs with raw meat or non treated garbage
- Control of rats in pig farms

Tuberculosis

Wednesday, December 25, 2024

1:12 PM

TB > 10 million cases every year. every year 1.5... death) - streptomycin - 1st drug - historically known by name: consumption & wasting disease - white plague - 1711 1865 - hereditary (Jean Antoine Consideri) 1882 - Robert Koch discovered the Bacteria. 1921 - Vaccine called BCG found (But there are also mgs sub the drug) - due to HIV 1990s it ↑ - In 1994 1st drug - belongs to genus mycobacteria - mycobacterium TB is primary cause of TB. even though other M.TBC can cause it. - primary affects lung but also other organs too! (Cerebra)	Staphylococcus got rapid d. time. mycobacteria Cell wall & high lipid content / mycolic acids (difficult to stain) - Catalase positive except nonpathogenic M. kansasii ionized (INH) resistant M. tuberculosis * All mycobacteria stain Acid fast @ Ziehl-Neelsen stain. <u>M. Tuberculosis</u> → Obligate aerobes nonsporulated slow generation time (15-20hrs) - withstand weak disinfectant / drying. - Not show true / ve stained! But Acid-fast bacteria (imp to dtd stain) - once stained, it retain primary stain (dye)
<u>Virulence mechanism</u> - do not possess classical bact. virulence (toxin / capsule / fibres) - However, due to cell wall nature! - Can survive in lung macrophage avoided activated macrophage - Cell wall prevents host inflammation - Survive in lesion for long (asymptomatic Bacteria)	* Catalase positive except nonpathogenic M. kansasii ionized (INH) resistant M. tuberculosis * All mycobacteria stain Acid fast @ Ziehl-Neelsen stain. <u>M. Tuberculosis</u> → Obligate aerobes nonsporulated slow generation time (15-20hrs) - withstand weak disinfectant / drying. - Not show true / ve stained! But Acid-fast bacteria (imp to dtd stain) - once stained, it retain primary stain (dye)
<u>Risk Factor</u> - low socioeconomic status - Genetic disposition - Chronic infection, DM, liver damage (Es HIV = 10x @ risk to develop TB)	
Previous exposure to mycobacterial infection	

to develop TB)
 • Previous exposure to mycobacterial infection

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Sites of TB Disease

Pulmonary TB: in lungs (most TB cases) - No previous inf
 (2/3rd)

Extrapulmonary TB: such as Lymph, LNs, pleura, Bone, joints, liver, spleen, Brain, kidney (mediator)
 (1/3rd) (if HIV infected / person with no comp / Young children)

Miliary TB: disseminated (through B. stream) (rare)
 (90%)

Pulmonary Asymptomatic - infection arrested (unaware) - inactive TB. - clear
 * only 10% show Primary TB (via RT) - initial lesion in small bronchus or alveolus (mid lung periphery)

2 Courses

① lesion arrested: Tubercle undergo fibrosis & calcification.
 - latent.

② lesion break: caseous material discharged & cavity created (to facilitate spread via LN & B. stream).

not infectious?

TB @ extrapulmonary
 TB not infectious.

Progressive TB - resulting TB re expand / destruct tissue so causes

• Can undergo clinical symptoms. → Chronic Pneumonia
 hematogenous & lymphatic distribution. → tuberculous osteomyelitis
→ tuberculous meningitis
 - or severe case & miliary TB can occur.

Reactivation of TB: Caused by: Impaired immune status
 (ass'd malnutrition / alcohol / age advanced / stress / steroids / cancer)
 * Immuno suppressive medications / disease such as DM & AIDS are common preconditions.

→ Risk of Reactivation is via 10% every year.

Patho Physiology

Inhalation of bacteria got a chance.

→ Imm clearance of organs.

while Immuno Competent 2-23 of life time

- Not good test (less quality sample & sputum) for HIV patients

② cytology

* Use FNA / biopsy sample.

③ mycobacterial culture

→ Problem is by long incubation!

① semi-solid agar media

include bird break, 7H10 & 7H11 media

→ ingradient to enhance bacteria.

!!! to observe colony morphology (3-8 wks)

② sustainability test (selective media)

③ lowenstein Jensen

egg based media (solid media as select medium)

④ Broth media

to remy LJ-medium, middle break 7H9 & 7H12

→ 3-8 wks to grow

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MDR: Form of severe TB disease caused by m. TB resistant to @ least 2 most potent drugs anti-TB medicine (Rifampicin & Isoniazide)
→ Current Global Challenge with TB!

X DR + MDR TB plus resistance to any Fluoroquinolone & any one of 2nd line injectable class (amikacin & Kanamycin / Capreomycin)

TDR - TB = strain resistance to all 1st & 2nd line drugs with proven activity against m. TB.

* in 3 countries: India, Iran & Russia.

* documented but not by WHO.

Clinical presentation

* weight loss

* chest pain

* Night sweats

* Productive cough (> month)
• expectorant

* High fever

* Hemoptysis

Thin mucous purulent

high guafenesin

not drug (Bronchodilators
combination...)

Lab. Cont

→ Read this more

⇒ called it Genexpert

④ PCR & Gene amplification

If TST & IGRA are the way we say they are infected but not we say disease.

* the Adaptive (acquired) Response for measure

(Tuberculin Skin Test) : AKA Mantoux tuberculin skin test + to determine

If person was infected w/ TB bacteria.

Delayed hypersensitivity reaction

- w/ neg. tuberculin PPD

TST & neg. for neg. (2) 50% of HIV/AIDS

HIV/AIDS

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Current TB diagnosis : Study the table 7 D1 + memo and advantages & D2 advantages. (Page 33)

Treat $\div$ first line
 $\div$ 2nd line.

Prevention & Control

- early detection & treat
- ↓ overcrowding
- pasteurize milk → ↓ M. Bovis infection
- Immunization (BCG) - @ birth.

Multidrug Resistant TB

* Difficult to diagnose
→ time for culture

culture

only true subclinical
Active diagnosis

5. Putum $\Rightarrow$ is very low susceptible

except for ... 959322
P38 & 570 2542
P-9 + no 3716 and
021522.

Active disease
 Sputum $\Rightarrow$ is very low susceptible
 if ~~not~~ subclinical / Active 2nd 716...

① Latent: Positive / Positive / Negative / Negative / No inter
 TST Igra Culture Sputum smear
 no symptoms / Preventive therapy
 (BCG)

② Active: both positive / Positive / +ve / negative / Yes
 Yes mild to severe symptoms / culture / multidrug therapy

Leishmaniasis

Wednesday, December 25, 2024 1:18 PM

Leishmania

- * Parasitic change + dimorphic
- * Spirilla shaped in sandfly
- * ~~free~~ Promastigote / extracellular
- * ~~free~~ Amastigote found
- Kinetoplast only found in Leishmania & Trypanosomes

Kinetoplast: Compns extra nuclear DNA / modified mitochondrion

- DNA organized in multicopy - min & max circles.

Inside human: Amastigote
Inside sandfly: promastigote
Cutaneous + sandfly & ...

Leishmania spp causing disease

- ① L. mexicana complex
 - L. mexicana → **LCL, DCL**
 - L. Amazonensis → **LCL, DCL**
 - L. Venezuelensis → **LCL**
- ② L. Braziliensis complex
 - L. braziliensis → **LCL, MCL**
 - L. Peruviana → **LCL**
- ③ L. guyanensis complex
 - L. guyanensis → **LCL**
 - L. panamensis → **LCL**
- ④ L. denovani complex
 - L. denovani → **VL, MCL**
 - L. infantum / Chagasi → **VL, MCL**
- ⑤ L. major complex
 - L. major → **LCL**

Variety of disease manifestations
- focal distribution throughout world, tropics & subtropics
Newworld = Southern Texas to northern Argentina
* by Lutzomyia

Old world = Asia, Africa, W. mediterranean.

* by phlebotomus
genera of sand fly.

Clinical Spectrum of Leishmaniasis

- I. Cutaneous leishmaniasis (CL)
 - I. Localized - most common, benign & self-healing skin lesion
 - Oriental sore - AKA simple CL.
 - II. Diffused - rare cutaneous infection
 - Wound but - non ulcerated nodule
 - nodule - resembles Leptomatous
 - III - Mucocutaneous - simple skin lesion
 - metastasize to mucosa (nose & mouth rep.)
- II. Visceral leishmaniasis (VL)
 - AKA kala-azar → DARK SKIN
 - generalized out w/ RFS
 - high mortality
 - = PKDL (post-kala-azar dermal leishmaniasis)
 - skin manifestation
 - Caused by L. Donovani

Diagnosis of CL

- Dx + Skin lesion: chronic painless 'clean' ulcer
- nasopharyngeal lesions
- nodular lesions
- i Parasitology (smear & culture)
- * NNN-media (dermal scrapings, biopsy)
- i Delayed hypersensitivity: Leishman skin test

L. major → LCL only
 L. tropica complex → visceral! (all species) → LCL, mcl, antropt → LCL, mcl, antropt

Delayed hypersensitivity: Leishman skin test
 Serology: DAT, ELISA.

Visceral Leishmaniasis

- untreated - fatal (97%)
- Conventional dx: cumbersome
- Treatment toxic
- HIV Co-infection common
- epidemic potential.

Dx of VL

Clinical dx: Fever, hepatosplenomegaly, wt loss, pailor, lymphadenopathy, leucopenia

Parasitology: smear & culture (NNN) (SP, BM, LNA aspirate)

Delayed Hypersensitivity: Leishmania skin test

Serology: DAT, ELISA etc (RDTs)

Most common: Rk-39 RDT/DAT

dx culture: Promastigote (NNN)

Via smear + Amastigote (found inside macrophage) on peripheral blood smear.

Dx Challenges

- * inaccurate 40-60% clinical subset

recovery rates
 Spleen > liver > BM > LN

BM - @ iliac crest / sternum
 Spleen aspirate - quick procedure - but dangerous (bleeding, abscess)

NNN Cultures: Limitations

- * Contamination: bacteria & fungi
- * Incubation (7-10 days needed)
- * $T^{\circ} > 28^{\circ}C$ (domesticated)
- * subculture needed for BM.

@ Grady + Parasitology (Leishmania)

6+ is for >100,000 parasites per 1000 fields.

(see page 41)

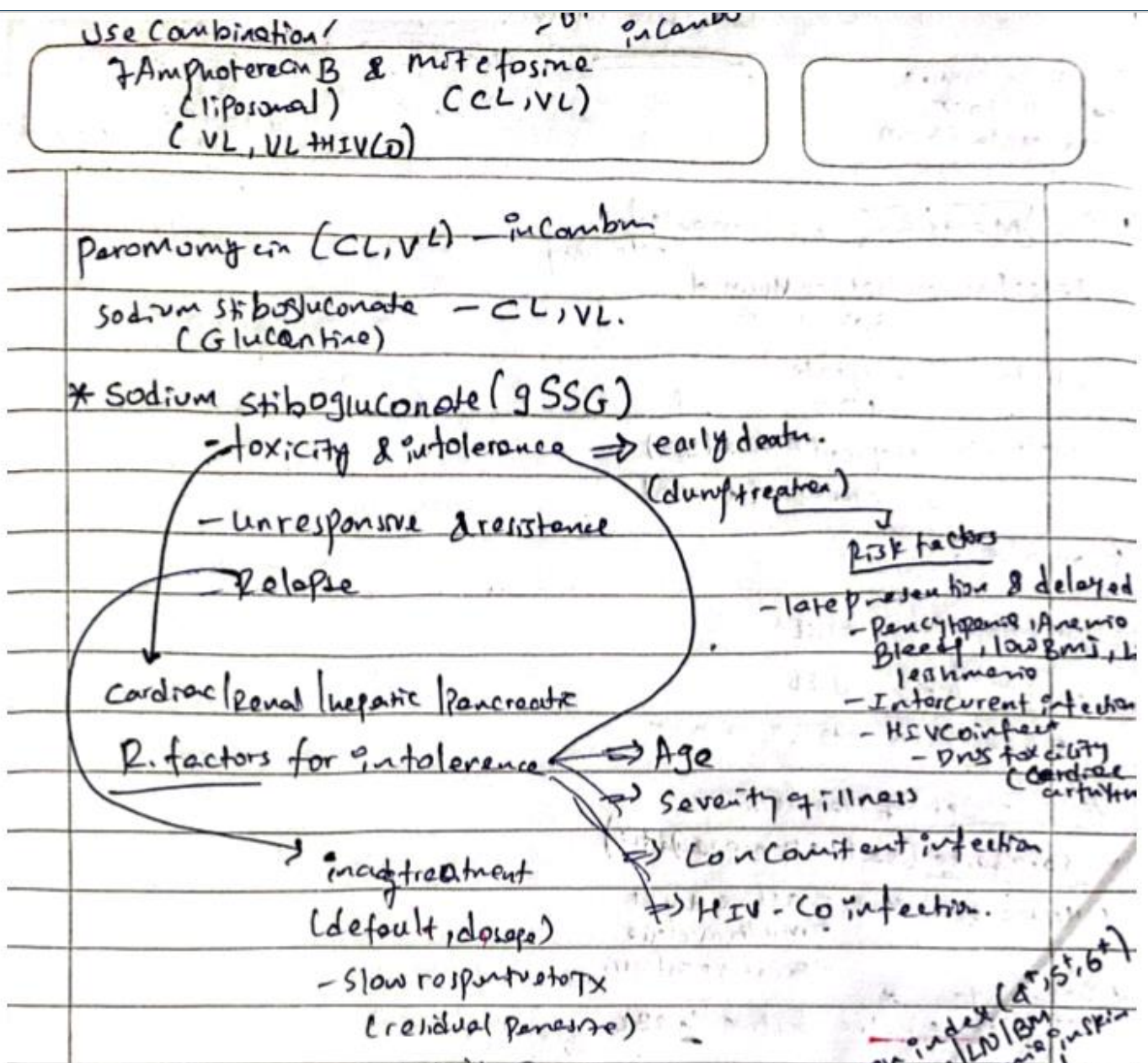
DAT: bit complicated (prohibitive cost)

- but good test
- high sensitivity & NPV
- low specificity & PPV
- Ad hoc / limited sup
- microhabitation st.

RDT: simple & speed (Rk-39) - High specificity & PPV - potential market

* inaccurate C ₄₀ - 60% clinical subset	8pk-39) - High specificity app
* parasitological (invasive/painful & risky technique)	- potential market - potential future dent.
* serological test : variable intrinsic validity	<u>Treatment</u>
* Skills : inadequate	First line : Sodium Stibogluconate 20 mg/kg, 30 days
	2nd line : Amphotericin B 1.0 mg (15 mg/kg)
	3rd line : Miltefosine 111 mg/day

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(residual parasite)

- Delayed Dx
- HIV Co-infection

High index (A)
Spleen/LD/BM
- parasitemia in skin
& blood

HIV Co-inf gVL ⇒ Atypical, clinical Parasitological & serological Presentation

Less frequent splenomegaly
PDXL w NO organomegaly!

- ve Rxn

* only available vaccine for Dogs.

Vector Control

animal reservoir

Env't capture

- vaccine

- enhance case detection & treatment

Remainder

* not 7 Ege

... read it please

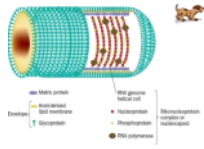
Rabies

Saturday, December 21, 2024 11:00 PM

Intro Rabies

- viral disease which causes encephalitis in warm blooded animals
- classical zoonotic infection

Causative agent (rabies virus)



- Family Rhabdoviridae, genus Lyssavirus

- Bullet shaped
- Enveloped
- Helical nucleocapsid
- Negative sense ss RNA

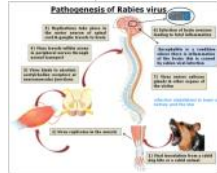
Source of infection

Domestic and wild animals

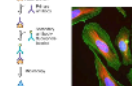
- Rabid dogs- 99% of human cases
- Rabid cats- 2-3%
- Vampire bats (America)

Transmission

- Common = rabid animal bite (saliva contains the virus)
- Animal scratch
- Rare
- Aerosol (for bat caves) or lab workers
- Person-to-person transmission -rare (organ transplantation)



Clinical representations

Period	Incubation period	Short prodromal stage	Acute neurologic phase
lasts	1-2 months/ 1-7 years (1% cases)	2-10 days	NB : acute ;fulminant , fatal encephalitis
Symptoms		Malaise , anorexia , headache	neurologic symptoms Cardiac arrest -- coma & death
		Photophobia , nausea & vomit Sore throat & fever	 CNS tissue examined for Negri bodies and rabies
Diagnosis		➢ RT-PCR from brain tissue ➢ Animal model (inoculating the inf...sample to suckling mice) what will confirm?	➢ Detection of rabies ag/ nucleic acid from tissue(IFT is rapid)  ➢ Serology -rabies Abs : IFT &ELISA (not detect till disease manifest) - Pathologic Diagnosis -Detection of Negri bodies in brain/spinal cord section • Pathognomonic of rabies(70 – 90% infected patients

Prevention and treatments

Preexposure prophylaxis

- vaccine administration

Postexposure prophylaxis

- Administering vaccine + VNA (viral neutralizing Abs i.e. human rabies

- what is the most source?
- What are the basic differences of the current vaccine : imovax rabies and the RABavert ?
- what contains the rabies virus?
- what are the common and rare transmissions?
- when is rabies fatal?
- what does the incubation period duration determined by? by the site of inoculations . on face is with short duration?
- cell culture : human embryo cell to see the cytopathic effects , list some of them.
- t is the most important ! and sensitive which test and how it carried out?
- what is the development in the history of rabies vaccine ?
- all are the cell culture vaccines.only to give for the human only!! Is it true?
- Explain the general features of rabies virus ?
- What does the rabies virus pathogenesis looks ?

immunoglobulin (HRIG - commonly used) or equine rabies immunoglobulin

Vaccines

1. inactivated vaccine for humans : several approved rabies vaccine strains adapted to various different cell types such as Human diploid cells (flurry (40-50 passages) , primary dog cell (flurry >200 passage) , kidney cells ,chick embryo cells (Paris Pasteur strain of rabbit fixed rabies) [see](#) page 20
2. live attenuated vaccine for pets

Administration of the rabies vaccine

	Preexposure prophylaxis	post exposure prophylaxis
For Who?	travelers to endemic areas Veterinarians Researchers	
	IM/ID injection of 1ml vaccine on days 0,7,21 &28 Boosting may be recommended (3-5 years interval)	IM injections on the Days 0,3,7,14 &30 . HRIG/ERIG Against viral glycoprotein G Unless recipient received previous vaccination against rabies !
categories of contact & recommended prophylaxis		

Current rabies vaccines

	IMOVAX	RabAvert
Produced by	Sanofi Pasteur Sa	Novartis vaccines & diagnostics
Obtained by	sterile , stable , freeze dried (freeze-dried vaccine) suspension prep from PM-1503-3M strain in infected human diploid cells ,MRC-5 strain	growing fixed virus Flury LEP(59th passage) in primary culture of chicken fibroblast
method	Concentrated (ultrafiltration) Inactivated by beta-propiolactone	Virus Inactivated by beta -propiolactone Proceeded (zonal centrifugation in sucrose density centrifuge)
Administration	IM ,Single dose ,no preservatives After reconstitution, Immediately give 1ml vaccine	IM , no preservatives Used Immediately after reconstitution with water for injection (WFI)

Rabies in Ethiopia

Species of animal associated with human rabies exposure

- Dog
- Cat
- Livestock
- Human
- Wild animals (fox, hyena, etc)

Factors that contribute for the occurrence of human rabies case in Ethiopia includes:-

- Traditional medicinal plants
- Spiritual beliefs
- Poor treatment seeking behavior due to various misperceptions
- Considering first aid as anti viral treatment

– vaccination in my country : rabies biologics (PEP) : Nerve tissue vaccines(NTV) widely used , NO RIG availability !

Preventions

- **Vaccinate pet animals- live attenuated vaccines**
- Keep your pets under direct supervision
- Remove any stray animals from your neighborhood.
- They may be unvaccinated and could be infected by the disease.
- Avoid direct contact with unfamiliar animals

[for deep dive in concepts](#)

1. Post-Exposure Prophylaxis (PEP)

Purpose: Administered after potential exposure to the rabies virus to prevent the onset of the disease.

Components:

Human Rabies Immune Globulin (HRIG): Provides immediate passive immunity by supplying antibodies. Type: An immune globulin preparation, not a vaccine.

Rabies Vaccine: Administered in a series of doses to stimulate the body's immune response and produce its own antibodies.

Administration Schedule:

HRIG: Given as soon as possible after exposure, infiltrated around the wound site.

accine: Typically given in four doses on days 0, 3, 7, and 14. A fifth dose on day 28 may be required for immunocompromised individuals.

Urgency: Must be started immediately after exposure to be effective.

Effectiveness: Highly effective when administered promptly and correctly.

Pre-Exposure Prophylaxis (PrEP)

Purpose: Administered before potential exposure to the rabies virus to provide protection in case of future exposure.

Components:

Rabies Vaccine: Administered in a series of doses to stimulate the immune system to produce antibodies.

Administration Schedule:

Three doses given on days 0, 7, and 21 or 28.

Target Groups:

People at high risk of exposure (e.g., veterinarians, animal handlers, travelers to endemic areas).

Booster Doses:

Periodic booster doses may be recommended for those at continued risk, determined by rabies antibody levels (titer).

Effectiveness: Provides long-term protection and reduces the need for HRIG if exposed.

Key Differences

Timing: PEP is administered after exposure, while PrEP is administered before exposure.

Components: PEP includes both HRIG and the rabies vaccine; PrEP only includes the rabies vaccine.

Urgency: PEP requires immediate administration after exposure; PrEP is given proactively to those at risk.

Key SimilaritiesGoal: Both aim to prevent rabies infection.

Vaccine Use: Both involve the administration of the rabies vaccine.

Effectiveness: Both are highly effective when administered according to guidelines

2. when administering rabies PEP, human rabies immune globulin (HRIG) should be given as close to the wound site as possible, but never at the same anatomical site as the first dose of the rabies vaccine. This is to ensure that the HRIG provides immediate antibodies directly at the site of the infection, while the vaccine stimulates the body's immune response¹.

If the wound is not accessible, the remaining HRIG can be administered intramuscularly at a site distant from where the rabies vaccine will be given. Typically, the vaccine is administered in the deltoid muscle of the arm, while HRIG is infiltrated into the wound and surrounding tissue

3. rabies PEP (Post-Exposure Prophylaxis) is unique for several reasons:

Immediate Protection: PEP includes the administration of human rabies immune globulin (HRIG), which provides immediate antibodies to neutralize the virus. This is crucial because rabies progresses rapidly once symptoms appear and is almost always fatal.

Combination of Passive and Active Immunization: PEP combines HRIG (passive immunization) with the rabies vaccine (active immunization). HRIG offers immediate protection, while the vaccine stimulates the body's immune system to produce its own antibodies over time².

Wound Care: Immediate and thorough wound cleansing is a critical part of rabies PEP. Cleaning the wound with soap and water, and sometimes a virucidal agent, can significantly reduce the risk of infection².

Vaccination Schedule: The rabies vaccine is administered in a series of four doses over a 14-day period. For immunocompromised individuals, a fifth dose is required¹.

No Contraindications for Pregnancy: Rabies PEP is safe to administer during pregnancy and does not require any changes to the regimen.

Effectiveness: When administered correctly and promptly, rabies PEP is highly effective in preventing the onset of rabies, even after exposure

4. Nb : PEP can be used for those HIV , Rabies tetanus , covid 19 , pertussis , and the Invasive Meningococcal Infection

The condition not due to microfilaria is onchocerca.

Onchocerca is a nodule formed under the skin due to the adult worm of Onchocerca volvulus

Leishmaniasis

- * Parasitic change + dimorphic
- * Spindle shaped in sandfly
- * ~~Amastigote~~ Promastigote / extracellular
- * ~~Amastigote~~ - Amastigote found
- * Carried intra cellular
- Kinetoplast only found in

Leishmania & Trypanosomes

- Kinetoplast:** Compounds extra nuclear DNA / modified mitochondrion
- DNA organised in multicircular & maxicircles.

- Inside human: Amastigote
- Inside sandfly: Promastigote
- spindle shape
- flagellated
- Cutaneous: Sandfly & ...

Leishmania spp causing disease

- ① L. mexicana complex
 - L. mexicana → LCL, DCL
 - L. azoanensis → LCL, DCL
 - L. venezuelensis → LCL
- ② L. braziliensis complex
 - L. braziliensis → LCL, MCL
 - L. peruviana → LCL
- ③ L. guyanaensis complex
 - L. guyanaensis → LCL
 - L. panamensis → LCL
- ④ L. denovani complex
 - L. denovani → VL, MCL
 - L. infantum / Chagasi → VL, MCL
- ⑤ L. major complex
 - L. major → LCL
- ⑥ L. tropica complex
 - L. tropica → LCL
 - L. tropica → LCL, MCL

Variety of Leishmania spp. - focal distribution throughout world, tropics & subtropics New World = Southern Texas to Northern Argentina

* by Lutzomyia

Old world = Asia, Africa, M. ea. Mediterranean

* by Phlebotomus genus of sand fly

Clinical Spectrum of Leishmaniasis

- I. Cutaneous leishmaniasis (CL)
 - I. Localized - most common, benign & self-healing skin lesion
 - Oriental sore - AKA simple CL - ulcerative
 - II. Diffused - rare cutaneous infection
 - Wound - non-ulcerated nodule
 - nodules - resembles Leptomonas
 - III - mucocutaneous - simple skin lesions
 - metastases to mucosa (nose & mouth)

- II. Visceral leishmaniasis (VL)
 - AKA kala-azar → DARK SKIN
 - generalized infection
 - high mortality
 - = PKDL (post-kala-azar dermal leishmaniasis)
 - skin manifestation
 - Caused by L. donovani

Diagnosis of CL

- Dx + skin lesion: chronic painless 'clean' ulcer
- nasopharyngeal lesions
- nodular lesions

Parasitology (smear & culture) * NNN-media (dermal scrapings)

Delayed hypersensitivity: Leishman skin test Serology: DAT, ELISA

Visceral Leishmaniasis

- untreated - fatal (95%)
- Conventional dx: cumbersome
- Treatment failure, ...
- HEV Co-infection common
- epidemic potential.

Dx of VL

Clinical dx: Fever, hepatosplenomegaly, wt loss, pancytopenia, lymphadenopathy, leukopenia

Parasitology: smear & culture (NAN)

(SP, BM, LN aspirate)

delayed hypersensitivity: leishmanin skin test

Serology: DAT, ELISA etc (RDTs)

most common: Rk-39 RDT/DAT

2a Culture + Promastigote

via smear + Amastigote (found inside macrophages)
~~on peripheral blood smear~~

Dx Challenges

- * inaccurate 60-60% clinical subset
- * parasitological (invasive/painful & risky technique)
- * serological test: variable intrinsic validity
- * SKINs: inadequate

recovery rates
Spleen > liver > BM > LN

BM - @ iliac crest / sternum
spleen aspirate - quick procedure
- but dangerous (bleeding, etc.)
d.b.m.c. (soft)

NAN cultures: limitations

- * contamination: bacteria & fungi
- * Incubation (7-10 days needed)
- * $T^0 > 28^{\circ}\text{C}$ (domestic)
- * subculture needed for BM.

@ gradif + Parasitology (Leishmania)

6+ for 100,000 parasites per 1000 fields.

!!! (see page 41)

!!! (prohibitive cost)

DAT: bit complicated
- but good test
- high sensitivity & NPV
- low ... & PPV
- Ad hoc / limited up
- microhabitation 57.

RDT: simple & speed
(Rk-39) - High specificity & PPV
- potential market
- potential for future

Treatment

First line: Sodium Stibogluconate
20 mg/kg, 30 days

2nd line: Amphotericin B
1.0 mg/kg (15 mg/1.5L)

3rd line: miltefosine
11 mg/kg

Use Combination!

Amphoterecin B & metefosine
(liposomal) (CL, VL)
(VL, VL + HIV co)

Paromomycin (CL, VL) - in comb

sodium stibogluconate - CL, VL.
(Glucantime)

* Sodium stibogluconate (SSG)

- toxicity & intolerance $\Rightarrow$ early death.

- unresponsive & resistance

- Relapse

Cardiac/renal/hepatic/pancreatic

R. factors for intolerance

$\Rightarrow$ Age

$\Rightarrow$ Severity of illness

$\Rightarrow$ Co-infectious infection

$\Rightarrow$ HIV-Co-infection.

$\Rightarrow$ Inadequate treatment

(default, relapse)

- slow respiratory

(residual parasite)

- Delayed Dx

- HIV Co-infection

HIV count $\geq 10^6$

$\Rightarrow$ Atypical, clinical

Parasitological & serological
Presentation

Low frequent splenomegaly
PdxL w no organomegaly!

Reminder

* not 7 Ego

read it please

* only available vaccine for Dogs.

Vector control

Animal reservoir

Env't appearance

- vaccine

- enhance case detection & treatment

Risk factors

- late presentation & delayed

- Pancytopenia (Anemia)

- Bleeding, low Bmi, L

- Intercurrent infection

- HIV Co-infection

- Dns toxicity

- Cardiac

- articular

- High index (4, 5, 6)

- Splenomegaly

- Parasitological

- blood

- ve Rx

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River Blindness : By Black fly vector

1. Causes
 - disturbance
2. Host
 - Human (Definite)
3. Caused by
 - O. Volvulus
4. Epidemiology
 - Visual Impairment (Not blind) complete
5. How is epidemiology in Ethiopia
 - Western part of Ethiopia
 - enormous economic crisis
 - itching & lack of sleep
6. Morphology
 - slender & blunt, coiled
 - Worms together in subcutaneous
 - Female size > male size
 - Adult vector is in the river
 - are blood feeders
7. Vector
 - Simulium damnosum
8. Why in fast flowing river
 - Concentration of O₂ for larvae more (PLP & CO₂ generated)
 - eggs attached to rocks
9. Penetration of larva in body
 - 10 yrs.
10.
 - Filariform @ thoracic region
 - ↳ infective
 - * larval stage do not enter to blood but to skin finally to eye & intraocular tissue
 - ↳ infective larva & itching
 - 11. Clinical manifestations
 - Adults cause it! → causes nodules
 - Africa below waist line
 - Central & South America head & above
 - 12. Most disease causing agent is?
 - micro Filaria released from nodule
 - I. Dermatitis
 - A. Acute Papular
 - B. Chronic hyperpigment, less pruritic
 - C. Erythematous; intense pruritic
 - D. atrophic
 - E. Depigmentation
 - II. Lymphadenopathy
 - * micro flare invade
 - * inguinal & femoral area
 - III. micro flare invade eye
 - * why west Africa blind but ethio sudanese - obscured
 - Savanna & Western Africa state

Part II Lymphatic Filariasis

14. Dx

* Slit lamp examination + free microfilariae visible to anterior chamber

1. The closely Related

B. malays → Indian East
B. timore
W. bancrofti - common

15. Laboratory diagnosis

* Skin snip, no need of blood!
You see 220-360mm Microfilariae

2. Vector.

* Mosquito

3. Unlike D. cercariae, it is PMS & daytime. Peripheral resides under lung, @ night, blood. adult - lymphatic vessel.

16. Dx of Adult

* Mazzotti test

* Serology, PCR.

4. Natural No reservoir host

• W. Bancrofti

17. Treatment

Ivermectin (reduces microfilar burden)

18. Surgical Cure, removal of subcutaneous nodule

Can be curative.

* Lymphedema & elephantiasis (adherent fibrosis tissue is removed)

5. Epidemiology

@ risk =

6. Ethiopia Case

Treatment : 7 Yrs.

19. Adult Vulvulus location.

On Bony Prominent area the adult can be found.

7. Adult Vs microfilariae

- 5-10 Yrs.

- No A. Canaan

- No nuclei

8. Cycle

→ @ night = microfilariae taken to blood.

4.

W. bancrofti = 10 pm - 2 am

W. malayi = 8 pm - 4 am

5.

Theans to Support Nocturnal

Periodicity

- +ve chemotaxis

- ↑ CO₂ in blood.

6.

Clinical manifestation

2 types + L. Filariosis (Adult worm)

± occult filariosis

Stage: Asymptomatic amicrofilaria

Stage: Asymptomatic microfilaria

- circulating microfilaria

- night time.

Stage: Acute manifestation

- Fungi & bacteria.

CRs *

Stage + chronic

lasts 10-15 yrs-

- Hydrocele.

! - chyluria (lymph urine).

- elephantias fluid

Ab + 2/1000 + Breast, Arm.

7. Occult (cryptic) filariasis (TPE)

@s result + hyperresponsive 7 filar

8. Lymphoedema

oedema

Folds

stage I - reverse @ night

No skin fold

stage II - irreversible @ night

Stage III - Irreversible swell

- shallow fold.

- *

Stage IV. :

stage V: not reversible

deep fold

: Irregular skin.

stage VI :

stage VII - need support to move.

Di	Topic IV. Loa-Loa.
Adult: lymph node	1. Vs to oncocerciasis
Giemsa stain.	* No effect. (Not severe form)
Etiol: chyluria.	2. Worm
Treatment	round worm
Diethylcarbamazepine citrate	
(DEC)	3. Not in etiologia?
Surgical Intervention: most imp.	Yeah but West Africa.
there may be ulcer.	4. Vector.
Vector Control	Day time biting fly.
Prophylaxis.	5. Loasos clinical manifest
<u>Notes</u>	- Asymptomatic
	I. Calabar swelling
	Adult lives in man's
	subcutaneous tissue.
	(NB) ÷ Not in etiologia!

Topic IV: D. medinensis

(Guinea Worm)

1. Eliminated or eradicated?

It will be eradicated after
from
14r Ethiopia like Marpon
(convergence eradication)

2. Morphology

Adult female: longest Nematode
Painful.

- Blister (so they want to be in water there)
Adult worm releases the larval stage)
into H₂O

3. Ways

* eating fish contaminated w...

4. DX

Worm out line under skin

5. Treatment

No medication.

slowly by stick comb and if
we cut it its infectious.

HAT & chagas

Wednesday, December 25, 2024 1:23 PM

Study Lecture: HAT & Chagas

1. 2 Trypanosoma species

- A. Stercoraria = T. cruzi (Latin A)
- B. salivaria = T. rhodesiense = T. gambiense

2. Which Protozoan w/ kinetoplast

* Leishmania

* Trypanosomiasis.

3. stages Trypanosome

HAT: Trypomastigote & epimastigote

Chagas: trypomastigote, epimastigote, Amastigote & promastigote

* NB: in Leishmania (Amastigote & prom)

4. stages in the cycle of Trypanosoma

* Tsetse fly: Epimastigote

* mammals: Trypomastigote (seen in blood)

5. Distribution of HAT

Western African: T. gambiense

Eastern African: T. rhodesiense (Zimbabwe)

6. Vector of HAT

Tsetse Fly - Glossina morsitans

7. Animal Reservoir of HAT

* T. rhodesiense: w/ animal

* T. gambiense = Anthroponosis (Human to human)
= Neotropical diseases (Pig/dog/cattle/sheep)
= Wild (Kob/ Hart's gazelle)

Clinical: Chancres w/ 1-2 cm
larger than Leishmania

* Winterbottom's sign of T. GAMBIENSE
(Lymph nodes @ Neck)

* Trypanosoma can have Neurological Complication: MYO edema

@ late stage + wasting skin disease.

8. Staging in HAT.

1st stage (Hemo-lymphatic stage)

* disseminate & proliferate of trypanosomes

* lymph & blood + fever
- Hepatosplenomegaly
- Lymphadenopathy

2nd stage: Meningo-encephalitic stage

* After weeks (T.b. r.) / months (T.b. g.)

* CNS invasion

Tsetse Fly - Glossina morsitans

Symptoms

- * CNS invasion
- Daytime
- Somnolence
- * Canx 2 death
- Nocturnal insomnia

(T69)

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CamScanner

2. How we diagnose.

- ① Blood = Buttercoat / thick film / capillary centrifugation
- ② CSF = lumbar puncture (Parasite WBC count)
- ③ L. Node = Aspiration - Aspiration

10. Treatment ?

Stage & species dependent.

* Drugs are both toxic & expensive.

* Current available are ÷ malar soprol, efloornitine, Niturtimox,
Pentemidide, Suramin

* drug combination

NECT (Niturtimox - efloornitine)

New Drug ÷ flexnidazole (Forty
coral)

11. Treatment in HAT.

Hemo lymphatic, T. rhodesiense = Suramin

Hemo lymphatic, T. gambiense = Suramin / flexnidazole

Meningo-encephalitic T. rhodesiense = malar soprol

meningo-encephalitic, T. gambiense = malar soprol &
flexnidazole

Topic II. CHAGAS Disease

1. Caused by

Topic 11. CHAGAS DISEASE

1. Caused by

trypanosome cruzi

discovered by Carlos Chagas

after his mentor Oswald Cruz

2 epidemiology

16-18 million infected = 100 million @ risk = 50,000 deaths
 * leads to heart & cardiac disease in 10-20%

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CS CamScanner

3. Vectors of Chagas
 "Bug"

triatomine bug
 reduviid bug
 assassin bug
 kissing bug
 cone nose bug

Common name

Genus
 * Triatoma
 & Rhodnius
 & Panstrongylus

4. Mode of transmission → feed on blood & defecate!

* Essentially v. born! (Triatomine bugs - Contaminated food)

* Transfusion - urban areas (in Bolivia!)
 ↳ essentially the promastigotes

* Congenital @ any stage of cruzi infection - Premature labor, laborious & neonatal death

* Accidental + ingestion food
 Contaminated by metacyclic trypomastigotes;
 lab accidents.

5. CHAGAS disease: disease stages

(a) acute phase: active infection

Duration = (1-4 months)

* most are asymptomatic but children are symptomatic

(b) Intermediate:

✓ life long problem.

* 10-30% of latency

- asymptomatic (no detectable parasitemia) / seropositive.

- asymptomatic (no detectable parasitemia) / seropositive.

(2) chronic phase:

* 10-30% of infected $\rightarrow$ cardiomyopathy / mega syndromes.
(GI syndrome)
of leishman

6. what are features of each disease stage.

note - Acute phase features (IP = 1-2 weeks)

high parasitism - local inflammation + Roman's sign.
in myofibrils = Chagoma

(amastigotes) - Symptoms: fever / malaise / lymphadenopathy /
hepatosplenomegaly / traumatic diarrhea.
- acute / often fatal myocarditis in children.

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CamScanner

- cardiomyopathy (cardiomegaly)

= mega viscera (esophagus, colon)

7. Dx

① Parasite detection

PCN
direct
stained blood smear
inoculation into mice
in vitro culture $\rightarrow$ like leishmania.
xenodiagnosis.

② serological tests:

hemagglutination
elisa
immunofluorescence
Complement fixation.

DAT for leishmania!

8. Treatment

acute stage $\div$ Nifurtimox (8-16 mg/kg/day, 60-90 days)

$\div$ benznidazole (5-7 mg/kg/day, 30-120 days)

Chronic stage = treat symptoms

Chronic stage = treat symptoms

Malaria class note

Wednesday, December 25, 2024 2:45 PM

Lecture MCQ Notes : 08/03/2017

1. What is the name malaria comes from
mal- area = bad-air.
2. Species of malaria & the %? (55700)
w/c one @ western.
3. w/c anopheles mosquito in ethi?
Common
A. arabian
4. Public importance
1.5 million 0959 @ Africa.
death
5. the economic effect of malaria
6. w/c one is more wide spread? what about ethi
* P. vivax
7. who is affected?
* pregnant women due to anaemia
* children < 5 yrs
* individual from malaria area
8. what are major health problems ethi
* malaria
* HIV
* TB
9. How malaria is distributed in ethiopia
* seasonal (sep-Nov) & April & May @ lowland areas
* unstable malaria distribution
10. 4 major eco-epidemiological status of malaria in country (upto 2008)
* > 2500 (tree) - central highland
* lowland - seasonal
11. Sporozoite
- no flagella, rigid.
12. why NOT the male took blood (but plant tissue)
- die the female needs blood for her embryo's dev't.
13. How is the malaria transmitted? (2 ways)

Malaria during pregnancy

- Low bw
- High parasitemia
- Anemia, severe malaria
- Preterm labor
- Death

14. life cycle of Anopheles mosquito (caus malaria)

! First @ liver (exo... schizogony)

- mosquito takes gametocyte.

! - dev it takes @ female mosquito.

- blood erythrocyte stage

- Human is indefinite.

sexual
sporogony

asexual
schizogony

15. why one cause infection?

* merozoite effect of RBC so clinical symptoms begins.

Gametocytes only for transmission do not cause prodromal crisis.

16. which one onto intestine of mosquito?

- kinocyte row

17. from infected mosquito to human?

* sporozoites - to cause infection in human.

but immune cells...

we call it exoerythrocytic cycle $\rightarrow$ to liver.

* so merozoite is here.

* hypnozoite stage? (ever de malar coe...) $\Rightarrow$ relapsing fever
(dormant in liver)

$\rightarrow$ when our immune cell is try to challenge
it the gametocytes develops.

18. @ donation not erythrocyte stage happens

(so starts from merozoites again)

20. hypnozoite stages?

@ 50% Ad 95%

21. what makes them differ?

- geographical

- microscopic & thick smear.

- periodicity

- severity

- No relapse in P. falciparum no hypnozoite stages.

22. types

A. Complicated

B. Uncomplicated. @ liver stage; fever; malaria; peroxysm

- ① - Chilling stage (1-2 hrs) - shivering, muscle aches, RBC rupture
- ② - febrile stage (1-4 hrs) - vomit, nausea, convulsion / Pnt / Pmt / Pmt
- ③ - Sweating / crisis stage (1-2 hrs) - high perspiration

merozoite goes to new RBC & waste product is cleared hence fever down

* every 3 days: P.O & P.Vivax (Benign Tertian malaria)

* every 4 days: P. malariae (Quartan malaria)

* every 3 days: P. falciparum (Malignant Tertian malaria)

Based on
Paroxysm

23. How is the mechanism of Paroxysm occurs?

24. How Relapsing occurs? after complete treatment due to ... relapse
Recrudescence? not finishing treatment
remain reticulocytes stages multiply...

25. Pattern of invasion of erythrocytes from one species.

P. malariae - favor old RBC

P. falciparum - all RBC high multiplicative so severe

P. ovale / P. Vivax - Reticulocytes

26. Pathogenesis.

- chill fever & rigor due to merozoites
- Cerebral hypoxia due to cerebral malaria
- anemia

27. Signs & symptoms of severe / complicated malaria.

- Coma
- Severe anemia (Hct 15%) $\rightarrow$ not Hgb ??
- Convulsion
- Hypoglycemia (Quinine not Pnt 4-20 days the parasite itself takes glucose...)
- Hyperparasitemia (>5% RBC)
- Pulmonary edema

20. Human cerebral malaria by *P. falciparum*

P. falciparum - 1 seizure coma $\rightarrow$ no 25 coma $\rightarrow$ for 4 dp?

brain histology: Petechial & ring haemorrhage, sludge of blood in small vessel.
anterior (highly stained)

28. How does *P. falciparum* pathogenesis explained by theories.

A.

single RBC blood cell affected by dlt parasitism

B. Immunological theory: Host Immune response

via microglial activation, astrocyte damage to cross BBB via INF γ and

eventually, liver cannot test but cerebral malaria is severe.

29. malaria & pregnancy.

- Pregnant @ risk - b/c there hemoglobin is low.

- at risk from mom & child.

- Mother @ first time

mom + HSV. $\left\{ \begin{array}{l} \uparrow \text{Parasitaemia} \rightarrow \text{ever malaria} \\ \text{preterm born \& death} \end{array} \right.$

30. fetal complication of malaria

pregnancy for first time

why? is there any trimester?

31. Vascularization of placenta: $\downarrow$ O₂ for fetus.

31.

Species + specific + - strain specific. All *Plasmodium* species...

- Short term only

- Partial protection

severe manifestation + antit time.

32. How are immunity to malaria.

Congenital

- Duffy-ve RBC

- Thalassemia

- sick cell anemia

selective

- ovalocytosis

- G6PD

P. Vivax in West Africa

3. Diagnosis.

① Clinical Presumption
* afebrile child.

②

③ microscopy

Giemsa stain + Gold standard. (definitive diagnosis)

Advantage: to differentiate species.

Assessing severity of disease by measuring PBC count.

- Quantification of Parasitodensity

- followup

→ focus!

3d. How do we clinically diagnose? when to take blood sampling?

↳ before treatment @ febrile stage
production.

* For species identification Use thick smear.

size of RBC infected comparison.

35. What a diff of thin & thick smear.

thin: Concentration method → treated w methanol alcohol etc

thick: Species identification

(RBC is intact) -

Getting -ve is possible
Needs high magnification → hear RBC is 1000x

!! 36. Distinct morphological features (Put it on study list!)

↳ thin cytoplasm → P. falciparum (@ ring stage)

↳ is there a size increment? (of RBC)

What? PV or PO

@ 100% oil immersion

thin film

NOT 10x

↳ more than 1 parasite per RBC

Banana shaped gametocyte
male gametocyte (common!!)

only @ gametocyte stage

↳ Marginal forms 7% @ ring stage / open ring

(typical identification)

ovalets vivax
vivax vs falciparum. pore → P. falciparum.

(Trophozoite)

prophogous prometocyte

(Gamete --) 43

for P. falciparum

A P. falciparum

has a deep pin

capitulum

↳ Van form at Trophozoite stage

P. ovale & distorted, oval shaped
P. malariae.

mixed infection is expected sometimes.

* Vivax: she -- -- vacuole, all species 4 level,
* P. malariae: highly distorted shaped (vanform).

37. KVI :

↳ Rapid & highly sensitive

do not quantify density of infection.
not differentiate past & new infection.

* For areas no health facility.

38. molecular Test?

- for research based

- differentiates species.

- potential cross contamination of sample.

39. How to treat? How do we select Treatment.

drug Ethiopia FQ + co-trimoxazole
25472 PM
drug.

Typhoid malaria
Weight & Blood count & glucose.
Pregnant

Oral therapy possible: Vomiting
possibly needed
in severe

Drug Rx:

- associative liver/renal tests.

40. Life cycle & drug target?

- which drug target at liver stage?

Antimalarial drug may be given.

↳ for chloroquine resistance → Quinine --- Prohibit??

↳ Primaquine plus chloroquine for Vivax, ovale & P. malariae.

41. How do we prevent it. In Ethiopia?

* If not gametocyte you can control it.

* Insecticide residual --- Spray, I - T - Net

* malaria early detection

42. Vector Control, Environmental mgts

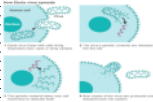
* spray only for 10% of population

* not to spray --

anthrax & brucellosis

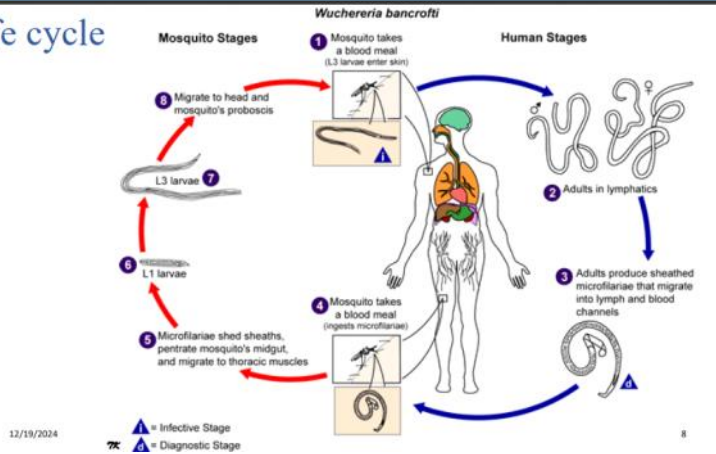
Wednesday, December 25, 2024 8:34 PM

similarity	Causative agents	Virulence factor	Transmission	Disease in Humans
✧ bioterrorism . ✧ —human to human is rare (not contagious) .				
Anthrax Caused by the spore-forming bacterium, Bacillus anthracis	• Rod-shaped • monomorphic • Endospore-forming, remain viable in soil and animal products for decades • Gram-positive • Aerobic respiration • Exotoxin producing • Grows best at 24-40°C • Incubation period: 1 to 15 days • Nonhemolytic • Nonmotile • Encapsulated (requires India Ink to visualize capsule) • Gram-positive, spore-forming rod	• Encoded on two plasmids • One encodes for the synthesis of a polyglutamic capsule (inhibits phagocytosis by macrophages) • Other bears genes encoding for the synthesis of exotoxins • !!! Exotoxins: three proteins • Protective antigen(PA) • Edema Factor (EF) • Lethal Factor (LF)	→ Human infection typically acquired through contact with anthrax-infected animals or ani Bacillus anthracis is transmitted mainly through the contact of infected animals or their products, and humans, resulting in cutaneous anthrax → It cannot be spread from human to human. → Biological Warfare Debate → animal products or atypically through intentional exposure	I .Cutaneous anthrax - anthracis comes into contact with a skin lesion, or cut. • The spores germinate, producing bacteria, which produce exotoxins to evade the macrophages • Infection is manifested in a painless ulcer with a necrotic (dead) center • Transmission: coming in contact with infected animals or their products through a skin lesion Symptoms of cutaneous anthrax • Boil-like skin lesion forming a painless ulcer • Swelling of the lymph glands • Fever • Headache II . Inhalation Anthrax • The spores are inhaled and lodge in the alveolar spaces • Alveolar macrophages engulf the spores. • Spores germinate within macrophages after failed phagocytosis • Bacteria proceed to lymph nodes and spread into bloodstream, where they begin to release the exotoxins • Transmission: inhaling the spores of B. anthracis Symptoms of Inhalation Anthrax • Initial symptoms: sore throat, mild fever, muscle aches • Severe difficulty breathing • Septic shock • Development of meningitis • Respiratory failure resulting in death III .Gastrointestinal Anthrax • Spores are consumed after eating undercooked meat • Spores can evade stomach acid due to their capsule • Once in the digestive tract, they begin to germinate and produce bacteria, which release exotoxins • After the macrophages have been lysed, the endosome enzymes begin degrading intestinal walls, allowing the bacteria to spread directly into the bloodstream • Transmission: Digesting undercooked meat containing spores Symptoms of Gastrointestinal anthrax • Serious gastrointestinal difficulty • Nausea • Fever • Abdominal pain • Vomiting of blood • Severe bloody diarrhea • Acute inflammation of intestinal tract • Loss of appetite
Brucella	✧ gram negative ✧ non-motile coccobacilli. ✧ In older cultures, irregular forms appear. ✧ They are arranged singly, in pairs, short chains or small clusters. ✧ They are nonsporing & no capsulated ✧ They are aerobic, but some strains require 5-10% carbon dioxide for primary isolation ✧ Fimbriae have not been described The Organism ✧ Multiple species – Associated with certain hosts ✧ Environmental persistence – Withstands drying – Temperature, pH, humidity – Frozen and aborted materials, dust, soil		✧ Ingestion –Raw milk, unpasteurized dairy products –Rarely through undercooked meat ✧ Mucous membrane or abraded skin contact with infected tissues –Animal abortion products ✧ Vaginal discharge, aborted fetuses, placentas Transmission in Humans ✧ Aerosol – Laboratory, abattoirs – Pens, stables, slaughter houses ✧ Inoculation with vaccines – B. abortus strain 19, RB-51 – B. melitensis Rev-1 –Conjunctival splashes, injection ✧ Person-to-person transmission rare – rare cases due to blood transfusion, bone marrow transplantation and sexual transmission have been reported	Brucellosis is a systemic disease in which any organ or body system can be involved. • The incubation period varies between 1 and 5 weeks • The symptoms of brucellosis are nonspecific • The one common sign in all patients is an intermittent/irregular fever of variable duration, thus the term undulant fever Disease in Humans • Brucella infection may be asymptomatic or symptomatic - The disease is arbitrarily classified • acute (less than 8 weeks) • subacute (from 8 to 52 weeks) • chronic (more than 1 year) Complications of Brucellosis • Most common –Arthritis, spondylitis, epididymo-orchitis, chronic fatigue • Neurological – 5% of cases • Other – Ocular, cardiovascular, additional organs and tissue

	as it relates to	the virus	stitchy / pinocytosis / phagocytosis	source of infection & epidemiology	ways of transmission	pathogenesis	Clinical disease
Ebola	common	Ebola virus • Filovirus • Single stranded, negative sense, helical capsid, spiral • Inside cytoplasm - filamentous • Enveloped • Fuses hemorrhagic, petechial rash, organ failure • Hypovolemic shock • Monkey (but not primate) • 1 a class A bioterrorism agent known to cause highly lethal hemorrhagic fever		• Fruit bats - major cause of the transmission (live in regions of Africa that include areas of the outbreak (south eastern Guinea), eaten by people in central Africa and may play a key role in transmitting Ebola to apes and humans.  • Source of infection & epidemiology • Human-to-human • At-risk groups • Endemic in West Africa - Nigeria, Liberia, Sierra Leone, and Guinea	Infected bats to monkeys and apes, then humans infected while killing or butchering these animals. • Majority of people contract the virus through direct exposure to the body fluids of an infected person (such as saliva, blood, sweat, semen) also contaminated objects like syringes 		• Fever • Headache • Nausea • Muscular pain • Rash • Loss of consciousness, Seizures • Internal & External Bleeding • Massive internal bleeding leads to brain damage • Average case fatality rate is around 50% • Case fatality rates have varied from 25-90% in past outbreaks
Lassa virus	severe infection	1. lymphocytic choriomeningitis virus (LCMV) - an arenavirus general characteristics: Arenavirus • Sandy appearance • Black virus • Biconcave/ingraving virus (pathogenic) • Circular RNA • 2 segments of RNA (GP-shuffle) - antigenic shift • Helical capsid • Enveloped with plasma membrane of host cell - fragile 2. Lassa fever Lassa fever: encephalitis (Lassa fever) • Spread by rodents • Rash & hemorrhage from eyes • Fever (mouth also non-specific symptoms shift) • Diagnosis?	 	• Source of infection & epidemiology • Human-to-human • At-risk groups • Endemic in West Africa - Nigeria, Liberia, Sierra Leone, and Guinea		Endothelial cell damage/capillary leak • Platelet dysfunction • Impaired cardiac function • Cytokines and other soluble mediators of shock and inflammation 	Lassa fever • Incubation period of 5-21 days • Gradual onset of fever, headache, malaise, and other non-specific signs and symptoms • Pharyngitis, myalgia, retro-orbital pain, cough, and gastro-intestinal symptoms typically seen • A minority present with classic symptoms of bleeding, rash/facial swelling, and shock • Case fatality of hospitalized cases: 15-20% • Particularly severe in pregnant women and their offspring (on the plus think his wife was there... I don't know) • Deafness is a common sequelae 

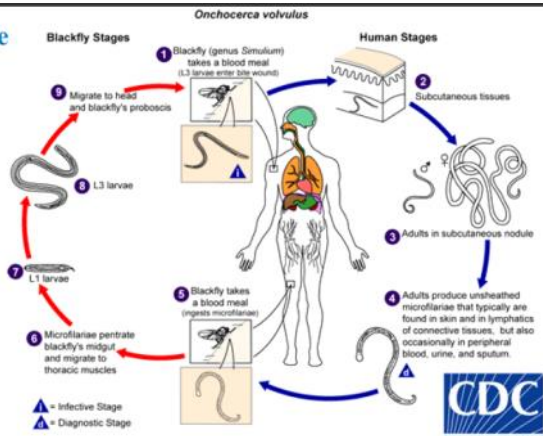
Lymphatic filariasis : no school kid .

Life cycle

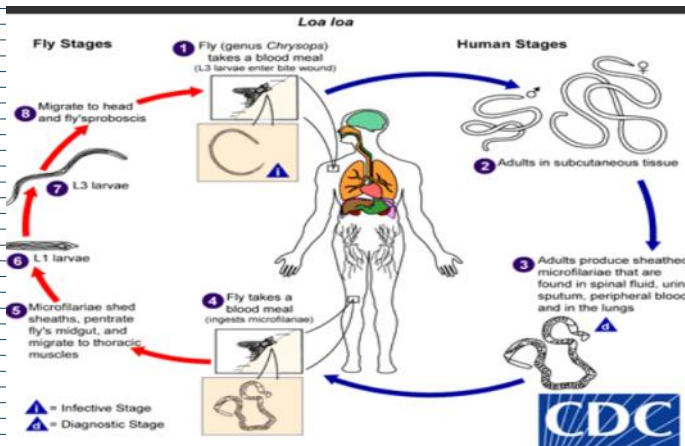


Onchocerciasis 'River blindness' = THE black fly & river man

Life Cycle



Loa-loa = eye worm

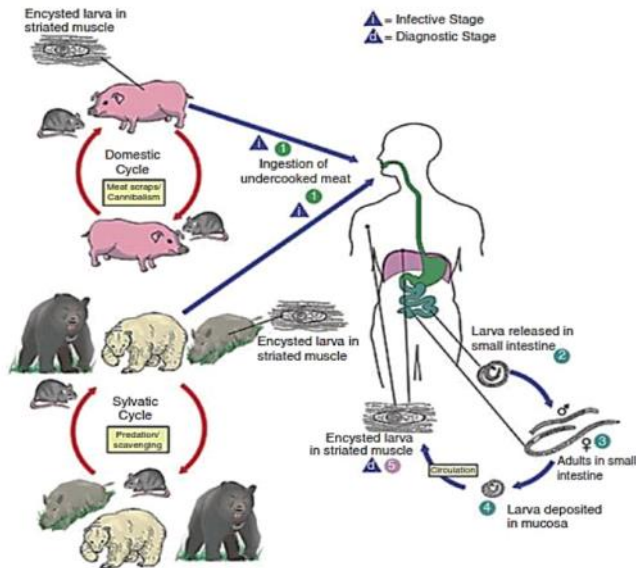


Schistosomiasis = the cloth washer

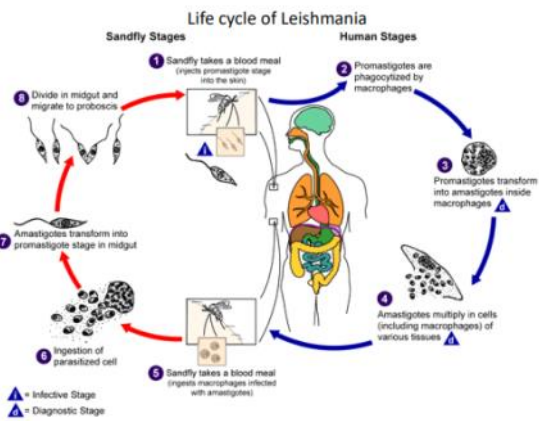
Fasciolosis = the animal farm

Hyatidosis = the sheep & the dog

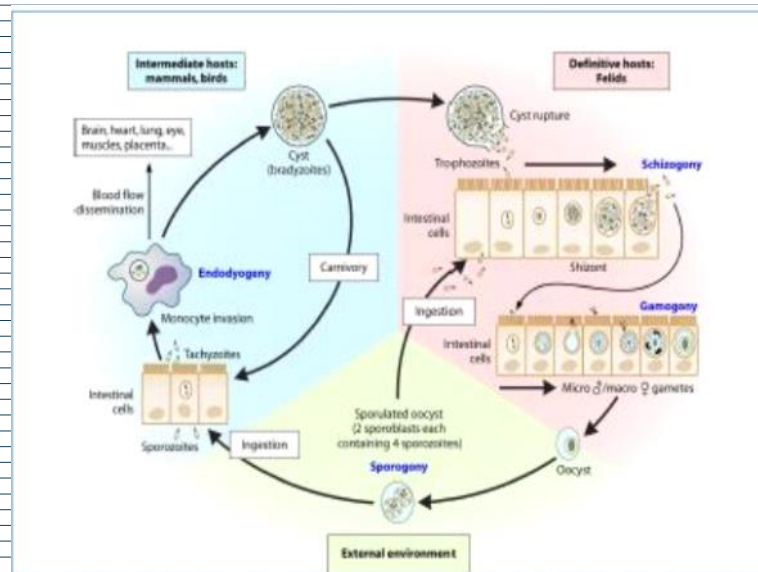
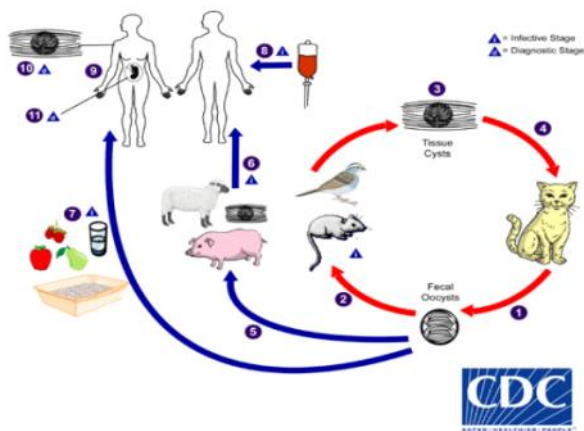
Trichinosis = the pig



Leishmaniasis



Toxoplasmosis



Fungus

Friday, December 27, 2024

2:23 AM

TB (labatory practice) 7

- * Primary Pathogen vs opp infection.
- * Systemic (endemic) + both immune competent & immune compromised
- Chester is def!
- * All dimorphic
- * entry + inhale
- * All cause pulmonary infection.
- * NO transmission human/animal but env't cause.
- * Parasitic (Yeast form)
- * all assume two forms.

Histoplasma Capsulatum (pear shaped)

- * mold in nature @ 25°C
- * Septated (tightly coiled) → smooth, hyaline spherical spores
- * 2 condida (mold form) → macroconidia = tuberculate macroconidia
- * Yeast @ 37°C (Parasitic) → microconidia = small, round, slightly rough w/ spine projection
- oval body.
- white twin walls

Histoplasmosis.

- ↳ incidence varies.
- * Soil contaminated w/ bird / bats dropping.
- * Teleomorph: *A. jellomyces Capsulans*
- * uncellular, elliptical / pyriform (pear shaped)
- * spindle shaped
- * dubuski are indistinguishable.
- * how to identify microconidia & hyphal element.

teremip... / 14 sellom... capsule... shable.

* How? Inhaled microconidia & hyphal fragment.

* 95% asymptomatic

* 100% attack rate (immuno compromised)

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Pathogenesis

TRP mechanism to podcer

for fungal infection: cell mediated defense mech Immunity!

So suppressed immunity, severe

Steps inhale microconidia (Primary cutaneous inoculation)

↓
Convert to budding yeast (in sac)

↓
Phagocytosis by alveolar macrophages

↓
Restriction of Growth to RE's by b. sm

↓
Suppression of cell mediated immunity

! Clinical forms (Human)

(A) Pulmonary (acute/chronic)

Acute

- lesion in lung's nodes.

* fever / chest pain / ill feeling / dry cough

Chronic rare (1/100,000)

③ disseminated.

Chronic Rare (1/100,000)

* mediastinal
granulomas & fibrosis
* fibrosis & locking
of mediastinal
structure

* Pre existing
structural
lung defect
ie COPD, emphyse

(less
obvious
to chest
clap)

* reactivation / re infection

* Chronic
pneumonia
or not incavates
(a sputum)

* it resembles TB
within 1 month! Men!

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B. disseminated.

* Immuno compromised
where H. capsulatum can spread from lungs into
other organs
(Infants & HIV)

↳ fever, malaise, Petechiae / skin lesions
(cutaneous histoplasmosis)

- mucos ulceration.

- enlarged lung / spleen / lymph node

* Acute disseminated histoplasmosis is fatal
within days to weeks if untreated.

Lab

Sample: sputum / blood / CSF / Bone marrow /
Skin scraping / biopsy / urine

Skin scraping / biopsy / urine

Direct exam + $\hookrightarrow$ Stains of fungi + Giemsa & Wright

• Intra & extracellular yeast cells.

Very specific

$\hookrightarrow$ Routine histology + H&E
Small yeast (2-4 μ m) intracellular
Yeast in macrophage
 $\hookrightarrow$ Sputum + KOH | Calcofluor stain

* Culture $\rightarrow$ mould @ 25 $^{\circ}$ C (tuberculose culture medium)
 $\div$ Converts to yeast on enriched medium (@ 37 $^{\circ}$ C)

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VARiants

$\rightarrow$ not doubled wall.

H. Capsulatum var. duboisii

* Disease: African histoplasmosis

Vs + classical histoplasmosis

- Indistinguishable in their mold forms. (same)
- differ in their parasitic form: yeast cells.
- Pronounced giant cell formed in infected tissue.
- $\downarrow$ (diminished) Pulmonary involvement

- ↓ (diminished) pulmonary involvement
(pulmonary lesion: uncommon)

- ↑ frequency of skin & bone lesions
(greater)

Inhalation of only
few *Arturoconidia*
produces
primary coccidio-
mycosis

most virulent of all
human mycotic pathogens

② *Coccidioides* Immitts

* 2 species *C. Immitts* & *C. posadasii*

* distinguished only by genetic analysis &
different rates of growth in
high salt concent.

C. posadasii grows more
slowly

* differs in geographic distribution & genome.

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Coccidioides morphological features

C. immitts (*C. posadasii*) = dimorphic fungus

* mold in nature / cultured @ 10-25°C

⇒ * Barrel shaped arthroconidia

* unique parasitic stage

its features are
complex

Tissue (37°C): spherules
↳ so genome

Tissue (37°C): spherules → so's genome.
 spherules (30-60 μm) - filled w
 endospores.
 & w/c are 3-5 μm
 in diameter.
 NB
 B/c
 the parasitic
 stage
 is not yeast

not common in Africa!
Coccidioides Immitis → compare it w the
 Kinetoplasma.

Disease: Coccidioidomycosis.

Epidemiology: Endemic (SW. United state, Mexico,
 Central & South America)
 : Central
 valley of (Valley fever)
 California

Source & transmission

- ① Natural reservoir (soil & dry / low rainfall areas)
- ② Inhalation of fungal spore.

* Coccidiomycosis is primarily : Pulmonary
 & sease.
 * 60% in endemic areas (asymptomatic). 2% suffer
 "Blue-like" illness

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Pathogenesis

* Inhalation of infectious Particle (Arthroconidia)

- * Inhalation of infectious particles - *Coccidia*
- * Sphule formed in vivo.
- * engulfment in phagosomes by alveolar MOs.
- * activation of macrophages (phagosome-lysosome fusion) → killing.
- ↓
- deposition leads to local inflammatory rxn.
- * Immunosuppression due to binding of complex to cells bearing FC receptors.
- * endospore released from sporules of *paratuberculosis*
- * inhaled as arthroconidia.
- * arthroconidia (as alternate cells undergo mycelium)

Clinical forms

- * Pulmonary coccidiosis
- * Disseminated coccidiosis
- * Primary coccidial pneumonia
- * Cutaneous (17% - 29% of all community-acquired pneumonia)

Dx samples: sputum, tissue, biopsy.

Treat by
• Amphotericin B
• Fluconazole
• Itraconazole

Direct exam: KOH & H & E

- Spherule in sputum, exudates, etc

!! * Preferred over Culture → highly infectious
nature of med.
specially C. Immitis

Culture

* Grows in most of media. (5% of sheep blood agar)
chocolate agar

S DA, myosel agar

as mold grows

Brain heart infusion
agar w/out blood

in 2-4 week

→ spherules

→ @ 37°C in tissue.

→ special medium: modified Converse lipid
medium (purified

agar & incubate
@ 40°C in candle
jar.

Safety

for known Coccioid

w/ 10% CO₂ for

Species level 3 Caution

3-5 days

Lab dx

& spherule.

thrombocytes each w/ at least 2 nucleoli / divided

... & spore.

thrombocytes (each is at least 2 nucleoli) driven
by the articulation of septate hyphae.

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CamScanner

3. Blastomycosis dermatitidis

↳ Systemic Fungal Int by B. dermatitidis
It called Blastomycosis. Drug

Morphological

* dimorphic fungus

- * Amphotericin B
- * Intracavitary
- * fluconazole
- * corrective surgery

* Broad-based budding yeasts (8-15 μ m
in diameter) Int > sue in culture
@ 37°C

- Distinctive

* Spherical hyaline 10-15 μ m

- multinucleated

* 'thick double Contoured' cells

* white to tan, filamentous / mold colony
on standard mycologic media @ 25°C

* mold form produces.

feature * round to oval / pear shaped
Conidia (2-10 μ m)

Epidemiology distribution: Africa, America & Asia (3 A's)

* Source (Natural reservoir) + Soil

AIB

* Transmission: Inhaled Conidia
In some case dogs also can have!

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Pathogenesis

* Inhaled → Infectious Particle | Primary Cutaneous inoculation

* infiltrated macrophages & neutrophils & Granuloma formation

* oxidative killing mechanism
neutrophils & fungicidal activity
of macrophage

Clinical

* Asymptomatic infection most of times

* Pulmonary infection

* Chronic Cutaneous infection

- Subcutaneous nodule & ulceration

* Disseminated infection

* Disseminated infection

(*) Skin, bone, gut; CNS, spleen

* Primary cutaneous infection

Lab dx: sputum & tissue

microscopy: KOH, H&E Yeast cells;

Culture: mold @ 25°C bud attached to parent via broad base.
* Conversion to yeast on enriched medium 37°C

go thru
we see
yeast
+ inside
unipolar bud
yeast line

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↳ Colonies on SDA @ 25°C Varies in morphology & growth

Group A vs Group B

* grow rapidly

* putty white mycelium

slowly grow as

glabrous, tan,

non sporulating colonies.

↳ Colonies on Blood agar @ 37°C

* wrinkled & folded

* glabrous & yeast like

* glabrous & Yeast like

It's Culture may be biohazard not as serious as C. Immits take care!

lab dx.

Serology: Immunodiff test
ELISA detect abs to
exoantigen - A.

Skin test: Blastomycin ag
↳ No diagnostic value
(lit ~~the~~ ref)

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CamScanner

Paracoccidioides Brasiliensis.

Disease: Paracoccidioidomycosis.

* systemic fungal infection caused by
P. brasiliensis

Morphological

Dimorphic fungi

bud attached
to

@ 20-40°C (human) → slender yeast (parent via

Dimorphic fungus

budding to

@ 37°C (tissue) : buddy yeast (parent v. g narrow base)

* Yeast like cells w double refractive wall /
& single / multiple buds (Blastoconidia)

↳ Variable size & no of Blastoconidia

(Pilot-wheel) morphology.

@ 25°C : hypha & conidia

* Gomori methamine Silver Stain.

- Showing multiple budding

Epidemiology

Attribution : Central & South America

* South American blastomycosis

Source / infection : ^{natural} reservoir

@ Humid air : 23°C - 28°C : inhalation of conidia

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CamScanner



Pathogenesis

* Fungus has protein in its cytoplasm
w/c binds only to estrogen NOT Testes
but

thus binds & converts to

but
Ctn.s bindy prevents Conversion to
Yeast form @ 37°C)

Cases in men = 15 time greater
age 30-50 yrs than woman.

* Yeast cell wall Polysaccharide (α -glucan)
stimulates Granuloma formation

Clinical

* Asymptomatic infection

* Latent form (Duration variable)

* Symptomatic infection * Nodular lesion in lung

* Dissemination to
other organs (rare!)

Affects men

Lab Dx

sample + Sputum Arose

sample + Sputum Arise

Microscopy: KOH & H&E multiple budding
yeast

Culture: Colonies (SDA) @ 25°C

- Slow growing variable

morphology

* may be flat, wrinkled &

* Serology

can be used

• Immunodiffusion

- Complement fixation

folded, glabrous,

swede-like downy texture

white to brownish at

then brown centres

* On BHI @ 37°C (mycelium
blood digar converts to
yeast

phase)

* Colonies white-tan
moist glabrous

become wrinkled

& heaped

↳ Got sort of Rote!

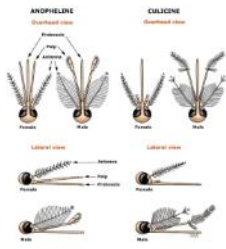
* Treat + Amphotericin B

Ketoconazole / Itraconazole

Sulphonamides.

Mosquitoes gallery

Thursday, December 26, 2024 10:33 AM

				Key for identification of sex of mosquito <table><tr><th>Structure</th><th>Male</th><th>Female</th></tr><tr><td>Antennae</td><td>Plumose</td><td>Pilose</td></tr><tr><td>Maxillary palps</td><td>As long as the proboscis with clubbed tips</td><td>As long as the proboscis with slender tips</td></tr><tr><td>Culicines</td><td>Longer than the proboscis with tilted tips</td><td>Very short</td></tr></table>				Structure	Male	Female	Antennae	Plumose	Pilose	Maxillary palps	As long as the proboscis with clubbed tips	As long as the proboscis with slender tips	Culicines	Longer than the proboscis with tilted tips	Very short	Key for identification of mosquito <table><tr><th>Stage</th><th>Anopheles</th><th>Culex</th><th>Aedes</th></tr><tr><td>Adult:</td><td></td><td></td><td></td></tr><tr><td>• Resting position</td><td>45° To the surface</td><td>Parallel</td><td>Parallel</td></tr><tr><td>• Color</td><td>Yellowish brown</td><td>Yellowish brown, non spotted leg and tyre absent</td><td>Black back ground with silvery markings on the back of thorax (lyre) & leg (spotted)</td></tr><tr><td>• Maxillary palps</td><td>As long as the proboscis with clubbed tip in the male As long as the proboscis with slender tip in the female</td><td>Longer than the proboscis in Male Very short in the female</td><td>Longer than the proboscis in Male Very short in the female</td></tr><tr><td>• Scutellum</td><td>One lobe with continous row of hairs</td><td>Trilobed with three bundles of hair</td><td>Trilobed with three bundles of hair</td></tr><tr><td>• Wings</td><td>Spotted specially the anterior margin</td><td>Not spotted</td><td>Not spotted</td></tr></table>				Stage	Anopheles	Culex	Aedes	Adult:				• Resting position	45° To the surface	Parallel	Parallel	• Color	Yellowish brown	Yellowish brown, non spotted leg and tyre absent	Black back ground with silvery markings on the back of thorax (lyre) & leg (spotted)	• Maxillary palps	As long as the proboscis with clubbed tip in the male As long as the proboscis with slender tip in the female	Longer than the proboscis in Male Very short in the female	Longer than the proboscis in Male Very short in the female	• Scutellum	One lobe with continous row of hairs	Trilobed with three bundles of hair	Trilobed with three bundles of hair	• Wings	Spotted specially the anterior margin	Not spotted	Not spotted	<table><tr><th>Stage</th><th>Anopheles</th><th>Culex</th><th>Aedes</th></tr><tr><td>Egg</td><td>Dark brown Slipper - shaped with air cells on either side Laid singly, making geometric patterns</td><td>Yellowish Ovoid with one end broader Laid in groups of 50 - 350 (egg raft)</td><td>Blackish Spindle - shaped w/ ornamented surface Laid singly</td></tr><tr><td>Larva:</td><td></td><td></td><td></td></tr><tr><td>• feeding & resting position</td><td>Parallel to water surface (floatation helped by palmate hairs)</td><td>Hang by siphonal tube making an angle to water surface</td><td>Hang by siphonal tube making an angle to water surface</td></tr><tr><td>• Palmate hairs</td><td>On dorsolateral surface of abdominal segments</td><td>Absent</td><td>Absent</td></tr><tr><td>• Respiratory siphon</td><td>Absent</td><td>Long and narrow cylindrical</td><td>Short and broad barrel shaped</td></tr><tr><td>• Respiratory spiracles</td><td>On dorsal surface of 8th abdominal segment</td><td>With 4 pairs of hair tufts at the end of siphon 4 rows on 8th segment</td><td>with one pair of hair tuft at the end of siphon One row on the 8th segment</td></tr><tr><td>• Comb</td><td>Absent</td><td></td><td></td></tr></table>				Stage	Anopheles	Culex	Aedes	Egg	Dark brown Slipper - shaped with air cells on either side Laid singly, making geometric patterns	Yellowish Ovoid with one end broader Laid in groups of 50 - 350 (egg raft)	Blackish Spindle - shaped w/ ornamented surface Laid singly	Larva:				• feeding & resting position	Parallel to water surface (floatation helped by palmate hairs)	Hang by siphonal tube making an angle to water surface	Hang by siphonal tube making an angle to water surface	• Palmate hairs	On dorsolateral surface of abdominal segments	Absent	Absent	• Respiratory siphon	Absent	Long and narrow cylindrical	Short and broad barrel shaped	• Respiratory spiracles	On dorsal surface of 8th abdominal segment	With 4 pairs of hair tufts at the end of siphon 4 rows on 8th segment	with one pair of hair tuft at the end of siphon One row on the 8th segment	• Comb	Absent		
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Diptera	Culicidae	Anopheles species	Anopheles mosquito																																																																																				
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		Culex species	Culex mosquito																																																																																				
	Glossinidae	Glossina species	Tsetse fly																																																																																				
	Phlebotomidae	Phlebotomine species	Sand fly																																																																																				
	Ceratopogonidae	Culicoides	Midges																																																																																				
	Simuliidae	Simulium species	Black flies																																																																																				
	Tabanidae	Tabanus species	Horse & Deer flies																																																																																				
	Muscidae	Musca domestica	House fly																																																																																				

thkds both feed on blood:s incomplete metamorphosis

Soft :Feeding is mainly nocturnal, feeds on many host thus known multi-host ticks

Hard ticks

•Rickettsial diseases (Rocky mountain spotted fever, Q fever)

•Spirochaetal diseases (Lyme disease)

•Viral diseases (Colorado tick fever, Encephalitis)

•Bacterial diseases (Tularemia)

•Protozoal diseases (Babesiosis)

Soft ticks

• Endemic relapsing fever (Borrelia duttoni)

•Q fever (Rickettsia burneti)

Scabies mites (Sarcoptidae)

• Caused by Sarcoptes scabiei, subclass Acari, family Sarcoptidae

• The female mites burrow into the upper layer of the skin appearing as tiny raised lines

Scabies is a common worldwide affects people of all races and social classes, spreads easily under crowded conditions

• Two genera of freshwater snails contain the intermediate hosts of mammalian schistosomes in Ethiopia.

• Biomphalaria: Schistosoma mansoni

• B. pfeifferi, B. sudanica, B. barthi

• Bulinus: S. hematobium & S. bovis

• B. tropicus, B. truncatus, B. hexaploides, B. octoploides,

B. abyscincus, B. africanus, B. forskalli, B. scalaris,

B. sericinus, B. reticulatus

Biomphalaria species

• Is a flattened pulmonate, planorbid snail with 3-7 whorls measuring 7-22 mm

• Four main groups are found in Africa:

• The pfeifferi –Africa and Middle East

• The Sudanica – East & West Africa

• The Choanomphala –Great lakes (Victoria)

• The Alexandria – North, East and South Africa

• B. glabrata –Western hemisphere

22

8

• Bulinus species

• is pulmonate snail with a left-handed opening

• Height varies 4-23 mm

• Two main groups of bulinid snails are important hosts:

• The Africanus group – Eastern and southern Africa and in most parts of south of Sahara.

• The truncatus, forskalli and reticulatus – Iran, Egypt,

Sudan, Madagascar & parts of East & West Africa

• Onchomelania is conical, 3-10mm and has 4-8 dextral whorls

Cockroaches have only three stages in their life cycle: egg, nymph and adults

emiptera - Bedbugs

• Two species of bedbug feed on humans:

• Cimex lectularius, most common globally

- Cimex hemipterus, occurs mainly in tropical countries

Fleas may be vectors of:

- Plague (Yersinia pestis) (in vomit) and flea-borne murine typhus (Rickettsia typhi) (in feces) (E.g. X. cheopis)
 - Chigoe or jigger flea (Tunga penetrans) female fleas 'burrows' into people's feet
 - Intermediate hosts of cestodes (Dipylidium caninum, Hymenolepis diminuta, Hymenolepis nana)
- Host/location: vibration, passing shadow, carbondioxide, certain chemicals
- Season: spring and autumn

Tunga penetrans (Sand flea, Chigger flea, Chigoe, or jigger flea): female embeds in skin

- Infestations is referred as Pediculosis inguinalis or phthiriasis
- Not known to transmit any disease under natural condition, but transmit louse-borne typhus in laboratory condition
- Their bites, injection of saliva and the deposition of feces
- Provoke itching, scratching, secondary infection, severe allergic reactions, pruritus
- Accompanied by purplish or blue stained lesions of epidermis without elevation (maculae caeruleae)

Phthiraptera (Lice – Sucking Lice)

- Are obligate blood feeders, flattened dorsoventrally, Wingless, Shortlegs
- Disease transmission: body louse
- Epidemic typhus, trench fever, epidemic relapsing fever, pediculosis (Vagabond's disease): infestation of lice

Myiasis more on child

house fly = Indirect mechanical transmission of microorganism (Enteric fever, yaws, diarrhea, Intestinal helminthiasis and protozoans, gastroenteritis, dysentery, anthrax, poliomyelitis, trachoma)

Study Lecture : Rabies & Newly emerging Virus.

Remembering notes

* In which case was egg embryonated @ emb? but for water
is in intestine.

* Questions (M-13-06 file)

* What are the hemorrhagic fever?

* transmission mechanism of P. f. & S. & late inf. dose

* opp pathogen

* severe form of tularemia

* mode of transmission & changes disease

* Kinetoplast ^{are} ~~containing~~ cellular organelles containing extranuclear DNA
(without correct abt.?)

* Disease staging in HAT & procedures.

* Leishmaniasis compare group & disease in V. & clinical presentation what is the abt.?

* HAT Caused by 2 species of trypanosome. what features

are species of disease caused by the trypanosome.

- Wilson's disease
- Winter but to name
- fever

* Compare & contrast life cycle of leishmaniasis & trypanosome.

Lecture Study * neurophysiology
do it & inc?

* when drugs were used for focal epilepsy? So treatment of epilepsy

caused by highly cysts can be treated

* where did I like skin tests!

* Prerequisite for leishmaniasis just up to 3/20

Monday 14/03/2026 ÷ questions

* determine stage & changes & Treatment why

* clinical & parasitological cases of leishmaniasis

as group of skin & mucous disease transmit to
low resistant body.

meloidosis & lecture Notes.

- similar to glandular disease
- epizootic (outbreak) disease ... gutch largely --
- Source + muddy water.

- Prolonged treatment.

- Biological weapon.

Cycle - 4 → Question survey

* Ectopic lesion of simian detection.

* Viral leucemia re-emerging virus considered as source of virus

* Virus & other reservoirs

* Ebola Virus disease

disease transmitted by

* body louse.

* Eschar is a sign of infection of the —

* Q fever is —

* MDR TB definition

* TB epidemiology infection type.

* Latent TB cases (features)

* Laboratory diagnosis of Anthrax disease.

Thursday : Questions Spokes

* Macfadden staining is used for —

* Agents of opportunistic mycoses —

* Candida albicans Transmission & Predisposing factors.

* Common isolated organism from mucormycosis pts

* Entomophthora mycosis

* Disadvantages of rapid Diagnostic test for malaria (RDT)

* Importance of thick blood smear preparation on test for malaria.

* Extracytic schizogony of *P. falciparum*

* events take place in RBC blood when malaria parasitism occurs

2: Questions

copies more in human diagnosis

plication of hybrid cyst

* Deuter cyst & Protoscoler hybrid cyst originated from

II M - Test I File II.

* Flavivirus is cross.

* Arbovirus types.

* Arbovirus cross

* Yellow fever, treatment/epidemiological agent/
metamorphosis

* Arboviral disease & their vectors.

* Brucellosis caused bacteria.

* LS media constituents.

* Borrelia & species.

* Borrelia escaping immune system mechanisms.

* Bartonella is that intracellular pathogen.

* exflagellation of Plasmodium results.

* Site of entry of the cycle of malaria.

Wednesday 1 DAY to Go.

Questions

- * Tools to control malaria
- * malaria during pregnancy complication.
responsible for
- * Relapse in P. vivax malaria.
- * Sensitive & specific diagnostic tool of malaria
- * Determinants of malaria outbreak.
- cycle 2
- * Species of malaria varies in what exs?
- * Nocturnal periodicity is evident in w. bancrofti
- * Portal entries of Coxiella burnetii.
- * Organism responsible for bubonic plague exs
- * rabies virus is contracted by what: saliva.
- * rabies exs infects, & IP, symptom duration, complie
- * effective treatments of rabies. - would clear
- immunity by modern cell
culture vax
- * forces driving an emerging virus. - HPI G intrav.
- * brucellosis disease exs def: , treatment & control
- * most accepted / Most used test for DA of brucellosis:
→ for confirmat.
- (NB) tube widal test has high specificity but
slide widal test has higher sensitivity.
→ for screening purpose.
- * Bacillus & Brucella bacteria common feature.
- * Likely r. of Antrax
cutaneous unless untreated.
- * About antrax virulence factors, is it common
forms of antrax.
- * mycobacteria. and exs, growth /
- * About latent TB infection / symptoms?
Do you feel sick
... ..

Basic form
We get while doing
the mqs oqgs...
Some repetitions you
get write on
your notes.
(sheep & dog)
(mane God)
* the most common
in blood test!

mcqs

* discoverer of TB

* Tubercle bacilli positive when.

* TB - risk of develop TB disease is highest in which years.

* dog bite & Gram -ve rods is for

* infections ways by Burkholderia pseudomallei

* organism which is commonly called the great mimicker is the Burkholderia pseudomallei.

* Subtype of Tularemia w lymphadenopathy but no ulcer. (ulceroglandular abscess)

* amp media to cultivate B. holderia pseudomallei from sterile site. - chocolate agar.

Cycle 3

* Tularemia - causative agent.
most difficult form

* ~~Scarcely~~ form of Tularemia. to recognise it.

* Brucellosis. ~~the~~ prevention.

* Anthrax lesion progression.

* inactive forms of anthrax.

* what does lethal factor results.

* Useful feature of Anthrax to be used as biological weapon.

* Inhabiting moist envt surface = Burkholderia pseudomallei.

* zoonosis of viral encephalomyelitis

* Case of Rabies (salivate, hallucinate, photophobia)

Quick Plans

* a virus noted for high incidence in ageneve dntz
can be isolated from most individual who have viral titres.

* tools to control malaria.

* microfilaria not in circulatory peripheral blood for 0. value

* Characteristics of plasmodium falciparum.

* Relapsing occur in cases of.

* main benefits of thin blood film.

mcgs

* Eschar is typical cr2 infection

vectors

Friday, December 27, 2024

3:12 AM

Arthropod : segmented

vs

Mollusks : unsegmented

* bio indicators in marine & freshwater

* Injects neurotoxic peptides (mollusks : Octopus, cone snails)

- what are harmful effects of Arthropods

- myiasis (infestation of fly larvae)

- How the Arthropods transfer Organisms to humans ?

via vomit drop / fecal pellet / saliva secretion

- Roles of Arthropods in H. Health

① Direct Cause of disease

- Ecto Parasitoses = Ticks
- Endo Parasitoses = myiasis flies
- Envenomization = wasp, bee, spider
- Allergic Rxn = Cockroach, dust mite
- Annoyance

② as Vectors (Host) of

pathogenic microbes
- flea
- mosquito

Vector → Biological (obligatory / natural vector)

mechanical

transmits

(NO reproduction or dev't change of pathogen)

① Propagative + No dev't but multiple

Eg: plague (flea)

② Cyclo Propagative = dev't + multiple

Eg: malaria (Anopheles)
+ Trypanosoma (tsetse)
Schistosoma (snail)

③ Cyclo dev't = dev't change

ES: filaria (Culex)

④ Transovarial = Infected parent

arthropod to offspring
(tick, mite)

Eg: Rickettsia
transmission

1st Arthropoda (5 class)
class insecta, Arachnide
& crustacea are imp
medically.

* only class insecta has

1-2 pairs of wings.

* No antenna in case of
Arachnide (octapoda)

* insecta includes: flies, bugs,
lice, fleas

* Arachnide includes: Ticks,
mites, scorpion.

* insecta = Head, thorax, Abdomen

General char of Arthropods

- * Arthropoda: jointed foot.
- * Invertebrate animal
 - segmented body
 - Bilateral symmetry
- pairs of jointed appendages
 - Rigid chitinous exoskeleton
 - Hemocoel system.

Primitive insect char

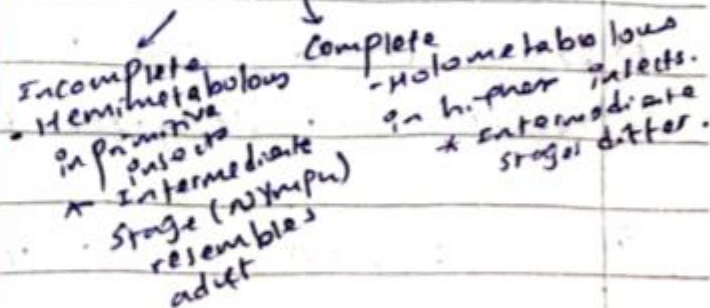
- Head = 6 fused segment
- Thorax = 3 segments
2 pair of appendage
- Abdomen = 12 segments
(No appendage)
- digestive system: foregut, mid
& hindgut.
- * circulatory system → Hemolymph

Hematophagous arthropods
host location strategies

1. Visual cues
2. odor cues
3. Heat cues

Notidiculous (notidiculous) life style.

Metamorphosis



Insecta

* Diptera (mosquito & fleas) → complete metamorphosis (Holomet)

• 2 wings / 2 pair.

• metamorphosis pair is balance, hatching (mechanism of metamorphosis)

* Nematocera (Mosquito, Sand fly, black fly)

* Brachycera (cherry sop)

* Cyclorrhapha (tsetse, house fly)

Mosquito (family of Culicidae)

W. wide except antarctica

* Toxorhynchitinae = don't feed blood (predators)

* Anopheline = blood feeders (eg. Anopheles)

* Culicinae = blood feeder (Culicines) eg. Culex, Aedes

- digestive system = foregut, mid & hindg.
- * circulatory sysm $\rightarrow$ Haemolymph in $\perp$ direction

* excretory syst $\rightarrow$ Malpighian tub

- * Culicinae = blood feeder (Culicines) Eg: Culex, Anopheles
- 4 larval stages.

Resting position

Spotting colour -

Adult mosquito - 10km

- Strong flier. ^{on antenna}
- Hair is scanty in female, many for male (Plumose) (Pilose)

- Head = kidney shaped compound eyes.

* mouth part collectively known as Proboscis

$\hookrightarrow$ saliva poured leads to hyperaemia & prevents blood coagulation. male feeds plant juice (but few like the female)

Thorax segment = Prothorax + mesothorax + metathorax

wings

Costa, subcosta are non branched veins

Abdomen

- 10 segment, 8 seen

- no scales in Anophelines

* fed mosquito - greatly extended abd. & looks oval red colour

egg stage - elongated oval

(Biggest - only)

@ dorsal site

Scutum

Scutellum

& post scutellum make it -

* Adult are non aquatic but others are aquatic

1. larva + head, thorax, abdomen

looks oval red colour
egg stage - elongated ovoid
 1. larva + head, thorax, abdomen
 aquatic
 (fused)
 * Respiratory spiracles
 * non feeding but nutritive, comma shaped stage
 * mosquito distinguished from others
 * Proboscis (visible forward projection) → suck / piercing type dependence
 * numerous Scales (on thorax, legs & wing ven)
 * fringe & scales along post. margin of wing

Blood feeding & gonotrophic cycle

* female got to take mosquito bite host & take blood to obtain nutrient for develop eggs in ovary
 * AKA autogenous dev 4
 Autogenous ^{sperm} flies → do not require blood
 * Spermatheca female fertilizes all eggs during her life time
 (not sterilized)

Scanned with
 CamScanner

* repeated process of feeding
 oogenesis & dev't.

Incycle (Gonotrophic cycle)

→ Half grown (Abdomen dark red anterior)
 → gravid (Abdomen dilated & whitish)
 Mid pit & blood chases through ovarian dev't

Oviposition & biology of eggs

eggs enter a state

- ① Quiescence → slowing down of certain process in response to env't condn. (Hatch if suitable)
- ② Diapause (physiological adaptation) → altered process to survive (until stimulus ends)

mosquito larva (distinguishing features) → Got prominent thorax than head & abdomen
 * legless, B. thorax (w/ head & abdomen)
 * & active larval instar (Needs wet mud)
 * head bearing well developed.

* larva respiration - spiracles (pair in mandibular & anal) & siphon (to xanthine & culicid)

* larva - respiration -> spiracles (pair in manson)
 - siphon (Toxorthychninae & Culicini)
 feeds: yeast, bacteria, protozoa & micros. decay
 - few are Carnivorous / cannibals. Plant
 in.

* larva is absent

from unintended water (rakes)

* large river / fast flow water.

W. 93 River b
 13.13.01
 10 year that

Pupae

* Non-feeding but moving

* aquatic & comma shaped.

* Head & --- fused to ?

Adult Biology

feed habit = Autrophagous
 Zoophagous

* or N. trophagous = On birds

biting habit = (day or night)

feed site = endo (endophagous) (exophagous)

rest site = endo (or outdoors)

Scanned with
 CamScanner

* Certain season time (autrophagous / endophagous / endophagous)

* at other time Predominantly zoophagous, exophagous or
 exophagous deep 55...

* female mosquito location

* stimuli -> breath (sweat)
 CO₂, lactic acid, odour

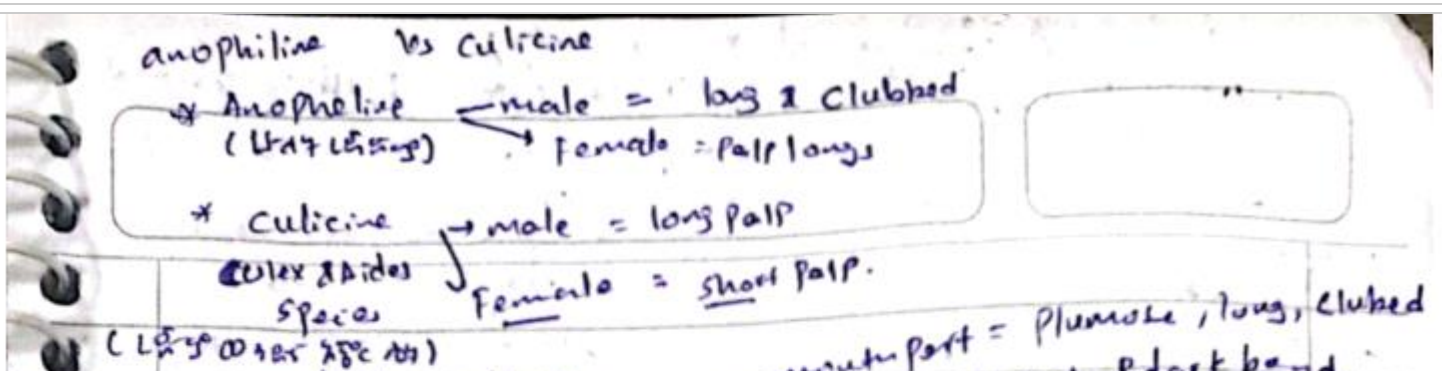
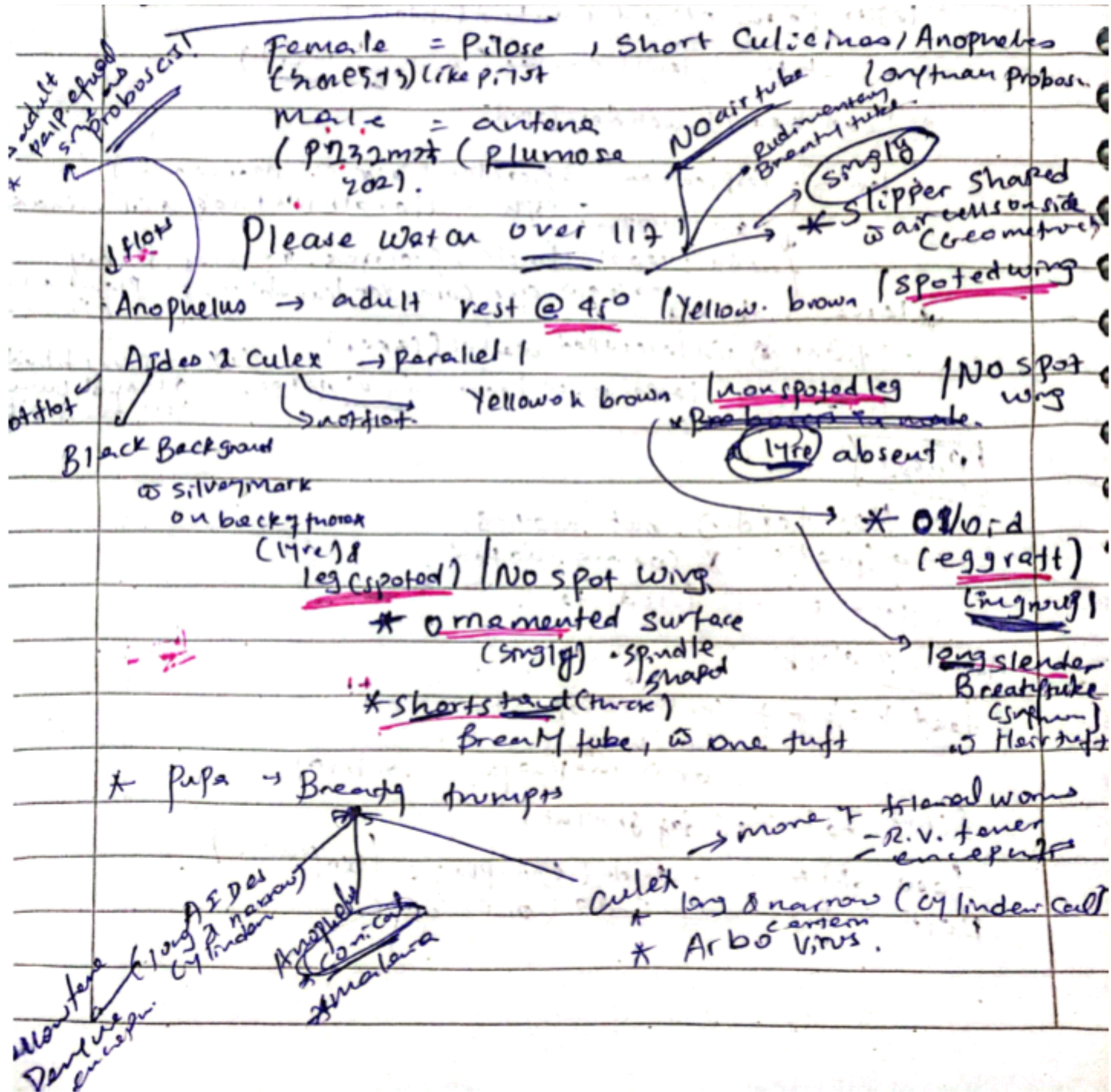
Body odor

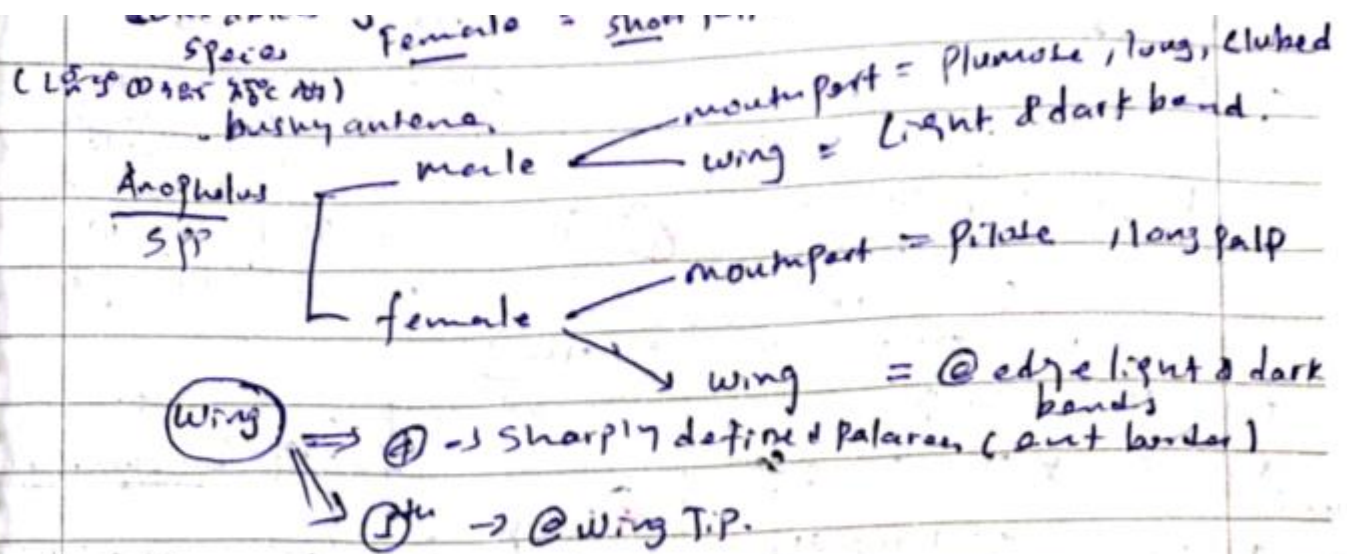
warmer

smell (Carnivorous)

Identification of sex mosquito

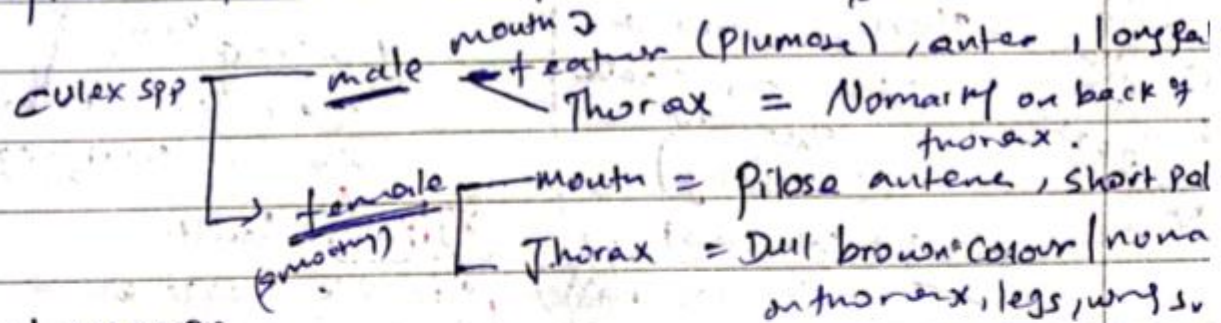
female = Pilose, Short Culicinas / Anopheles





x Conspicuous black & white scale -> pattern on thorax -> legs
got banded appearance. = Aedes.

x Dull brown Mosquitoes lacks prominent banding on legs or
conspicuous thoracic marking = Culex



Aedes aegypti

female. - Pilose, short palp

- 2 curved white lines (Y shape) on thorax

male

- feather long palp

- legs: black & white

- 2 curved white line thorax

- legs (black white band)

- Nematocera
- Simuliid (most Imp)
- Common in eth.

* economic & medical Importance

* Species in ethiopia

Found in Ethiopia:

- S. damnosum
- S. Neavei
- S. achraceum
- S. metallicum
- S. exsuum
- S. zonitum

* minute but stout

Female: Holoptic eye can

defend
on the
true

Female: Dichotic eye

Compound eyes.

Antenna No hair (short, stout)

mouth
part

short
broad
invariable

to the
segmented
nature
flap seen

* pull feeders (do not penetrate deep into tissue & lay o. ovipositor & skin & pupae)

* 3 pair leg + thorax + Abdomen = fine dark short hair (dorsally)

* wing + Coloured / short broad, no scale hair to enter into wet.

* better sexed view eye. (short hair abdo thick hairy, genitalia visible)

* black 418
egg → pupa → 1 pupa to 1 egg

(egg) ⇒ pale white 1 mm x 1.2 mm & 1d

* the smooth shell sticky sub 1d

- layed on flying water (rest & 1d on stone vegeta egg immersed objects)

* Communal Oviposition w thousand eggs. (Swarming nature)

Larva

Head = prominent part of feeding brushes (capitulum)

* Body = Got Proleg. w hooklets. (pseudopod), cigar shape

mature larva = blackish mark (911 spot)

* cocoon

pupa

* prominent filamentous, broad thin walled, respiratory gills.

* In group Age relate to colour we see (DARK means advance to adult)

Tips :: Adult

• female takes blood
* Swamp mating

* out door biting • S. damnosum (bimodal)

* biting region part.

• S. damnosum (legs)

• S. achraceum (head, thorax)

* bite same & never bites common in western Africa

* location & host

* Visual cue / sight (dark & light)

genitalia (visible)
 * @ resting -> closed flat
 ... body location ...

(*) location of host

* Visual cue (sight) (dark & ...)

* Odor

... 2-8 days

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[mcqs]

* Eschar is typical of infection

* larva got 4 pairs of legs which feeds

organic matter

* all segments include

head carries mouthparts

* pseudopods on Abdo

Egg -> larva -> pupa -> Adult (Complete)

Adult

* both drink sucrose

* plus the female feeds blood of human rodent / dogs / livestock / cats -

(*) biting @ nocturnal or (darkened room) @ day time

attracted to dark areas.

- most exophagous

(*) are weak fliers

in the (night) so can hopping flight

- dominant & ... for leishmaniasis

- Have red allergic bite of sand

* Papeter fever

* Ber to nellosa lora

Continued lecture

* repellent use.

* DEET

Sand fly

* leishmaniasis from 3 genera

(A) Phlebotomus

(B) Lutzomyia

(C) Sergentomyia - can bite but not disease

Important (feeds blood)

ergonomics but not disease
Causy.

Nature

- Sandy Color (pale)
- large black eye
- long & slitt like leg
- * all body dwms.
w dense long hairs
- singly layed eyes

* Bar tonellosis (Carri's disease)

Control & reservoirs
do 75 igarons...

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1 set fly (Glossinidae)

in tropical
regions.

- Identified by wings.

* female is ^{→ one larva @ time.} ~~laviporous~~
(do not lay eggs)

* Exophagic, both sexes feed
on blood.

Host location

* visual

* odor

(*) dark moving cue.

larva shaded @ water side
in soil.

* rest in humid dark sites.

pupa ± Barrel shaped w
2 post knobs.

* Complete metamorpho

Medical

Vectors of Trypano (animal & human)

Vectors of Trypano (animal & human)

animal sleep sickness (Nagana)

T. brucei brucei

* Human Sleep Sickness

T. B. gambiense

T. b. rhodesiense

(ST)

* Sterile male release method. ~~change body~~ place

- manual collection of pupae from soil

* Insecticide

* dark blue / black cloth

impregnated with pyrethrum

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Short Notes

* Ticks feed. @

2 segmented in larva $\Rightarrow$ 2 segment genome

* Gravid - ve.

Hanta: Rodent - human

Aerosol

* vascular endothelium (capillaries)

* Hemorrhagic

* detection: less specific

NB send your notes

as fire for the last moment only to see it!

leish

* endemic in West Africa

* Gradual onset of fever.

* can cause fetal death.

* lab based dx is important.

Tubercle dear fly, rabbit.

÷ true aerobic motile fastid.

÷ anti phago - lysosome

÷ most virulent (without being killed it's phagocytosed)!

* chocolate agar

Glander

Meloidosis

Glander

- * rare
- * Burkholderia mallei / zoon
- non motile

- donkey
* high fatal (95%)

* safety pin.

* long term Antibiotic.

Meloidosis

* Opportunistic

* It is end of air
(most lyf)

B. pseudomallei

Yellow fever

Serological is main way. (demonstrate Neutralising ---)

dengue

pc & Not serology b/c of cross linking!

Toxoplasmosis

Friday, December 27, 2024 4:19 AM

1. Migrate to head & 1/3 of body

Basic Mcg based study = Toxoplasmosis - Most common zoonotic

* only known DH = Members of Felidae (domestic cat & their relative)
(sexual cycle) → sexual reproduction takes place in intestinal epithelium of cat host & cysts shed in sporulated

* Shed from cat feces = unsporulated oocysts (only shed during the 2 week period of intestinal phase. (million) for 1-3 wks. of oocyst)

* oocyst (140 5 days) = Sporulated @ env't = Infective
* are major source of infection! * takes 48 hrs to sporulate! (So not as soon as dropped)

* IH (Natural) = Soil | H₂O | Plant Contaminated w/ OO cysts
↳ birds
↳ Rodents
↳ contains the Sporozoite form of Toxo. parasite
↳ once swallowed OO cyst burst in intestine! then sporozoite

* oocyst → Transformation → tachyzoite (shortly after injection)
NB: Found in the wild & domestic cat not other!

* Tachyzoite located = Neural / muscle tissue → develops to Bradyzo (tissue cyst)

* Cats got infected = Consuming Sporulated oocysts.
directly (ingestion) ↳

* Animals bred for longevity = ingestion of Sporulated oocysts (in env)
↳ infected by tissue cyst.
After

Human OO

trophozoites

Human ?

Cyst.

Citrophozoites

- Eating under cooked meat = tissue cyst
- Consume H₂O/food w/ cat feces / ^{contaminant} _{ent} samples (fecal contamination)
- Blood transfusion / Transplant ^{contaminated} _{donor} blood
- Transplacentally to fetus

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* Tissue cyst of parasite Formed in → Skeletal muscle, myocardium

(In human)

(NB)

Can remain throughout

Life of Host = Tissue cyst

Brain & eyes
_{Cerebral}

*

Dx = Serology

= Stained biopsy specimen (Tissue cyst)

Dx of congenital inf = detect T. gondii DNA

Classical triad

- * Intra cranial Calc In Amniotic fluid
- * Chorioretinitis
- * Hydrocephalus

(PCR)

Others

Still born
Psychosis

Clinical

* Immuno competent : Asymptomatic

: 10 - 20% With Acute infection

- Cervical L. Adenopathy

- flu like symptoms

: Benign & limited (wks - months)

: Ocular infection & Vision loss

(Rare)

→ Immunodeficient
↳ CNS disease

↳ Retinochoroiditis, Pneumonia, other systemic

In AIDS pts = Toxoplasmic encephalitis
(most C. Cause)

Intra cerebral mass lesion → via reactivation

Chronic
infective

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Dx observe parasite
(in BAL from immuno compromised Pt ILN dx biopsy)

⊗ Isolate parasite from blood / fluids (Intraperitoneal inoculation into mice / tissue culture.

6-10 days (post inoculation)

If no organisms serology (4-6 wks) post inoculation

- Obligate Intracellular (Coccidia. (Promastigote w/ no flagellum, spindle looks
- 3 forms 1. Tachyzoite (spindle) - rapid reproduce
- 2. Bradyzoite (slower ...) in tissue
- 3. Sporozoite (oocyst) ⇒ infective stage.

Tachyzoite → in vacuole body to multiply

- any tissue / organ

- @ acute phase (extraintestinal phase)

* Only in the intestinal phase

- @ acute phase (extraintestinal phase)
- ! * only cats for intestinal phase.
- (DH)

Bradyzoite

- 2-3 weeks after.

divide more slowly
protective membrane formed...
cysts (containing parasite cell) ^{called} Zoite cysts

→ contain cells called bradyzoites.

cyst = parasite cell + cell
(zoite cysts) (bradyzoite)

the cyst itself

contains the

in Brain, eye, heart & spleen
can stay longer or for life

Oocyst

CATS: Toxo infects S. Intestine → Reproduce asexual
few days later (rapid reproduction internally) → the
cells transform to Sexual form & enclosed
in cyst → oocyst

Chronic infection = bradyzoite
Acute infection = tachyzoite.

- * Sexual cycle : the oocyst produced.
- * Asexual cycle : ?

* env't → abt week oocyst mature.

Acquired

① Immuno competent Children = Postnatal acquired = Asymp

Systemic infection = lymphadenopathy, fever, Hepatosplenomegaly.

Immuno Compromised = Disseminated infection (myocarditis, Pneumonia)

Immuno Compromised = Disseminated Infection (myocarditis, pneumonia, CNS toxoplasmosis)

- Fever
- Stiff neck
- myalgia
- arthralgia
- rash (maculopapular)

Spared
Palm
Sole

others
symp

local: General L. adenopathy
hepatomegaly, hepatitis
reactive lymphocytosis, meningeal
abscess, encephalitis, confus
pneumonia, polyoma
retin.

- Chorio retinitis in 1%
(unilateral)

Trypanosomiasis (water buffalo fever)

Most manifestation = Cervical lymph node enlargement

Congenital toxoplasmosis.

- due to primary acquired maternal infection during gestation

* frequency of Transmission

⇒ to stage of

Severity of fetal damage
Transmission 5th trimester (1%)

Pregnancy
(when the
infection
occurs)

24th trimester
prophylaxis
& drug
more prolonged

- * 10% case (1st trimester)
- * 30% case (2nd trimester)
- * 60-70% case (3rd trimester)

* Placenta
resistance of BBB
as well as infection

Severity is inversely correlated.

1st trimester is more severe.

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* mental retarded

saquin
microcephalus / hydrocephalus / deafness / psychomotor] major sequela

* severe eye lesion @ early pregnancy

microphthalmia, cataract, intraocular pressure.

* severe eye lesion @ early pregnancy
 - microphthalmia, cataract, intraocular pressure.
 - strabismus, optic nerves, retinal neovascular.

UVeitis & retinoblastoma!
 * Can lead to blindness
 ↳ what like that river main w...

* Congenital Toxo

↳ main signs: Premature birth!

↳ Small GA
 ↳ retinal scar

↳ Jaundice

↳ mild thrombocytopenia

↳ CSF Pleocytosis

Severe signs

- Psychomotor retard

* ↓ IQ

- Seizure / Blind / Deafness

3 Triads

* cerebral Calcification

* Chorioretinitis

* hydrocephalus

Severe death

How is it Acquired?

Via oocyst = Sporozoites → Feco-oral
 (ingestion of con-
 sporia)

Via oyst = broodstock = raw meat / prey!

Via Tachyzoite = Transplacental (vertical trans-
 mission)

* Unpasteurized milk (Risk factor)

* Blood transfusion

Dx:

- Sabin-Feldman Dytest
 (DR) ⇒ oldest test
 ↳ type & serology

* Serology test (IgM & IgG)

* Diff. Agglutination
 Test (ACHS Test)

* IF.

initiates
 infection.

Treatment

1. Acquired Toxoplasmosis & Lymphadenopathy

• Do not require treatment unless severe.

* pyrimethamine

* sulfadiazine

* leucovorin.

Prevention

* Change litter box daily

* one vaccine for sheep

(no vaccine for human)

ternati
 A 27th man
 Chit evaluated
 in Child
 xclindamycin & pyrimethamine
 Co-trimoxazole therapy
 for HIV & co-trimoxazole
 as adjuvant therapy
 disease.

Congenital ---

↳ Newborn should be treated, whether clinical or not
 b/c treatment interrupts acute disease that damages
 vital organ.

↳ Infants treat for 1yr @ pyrimethamine, sulfadiazine
 & leucovorin

